

Readability Studio 2026

Readability Test Reference

~ READABILITY ANALYSIS SOFTWARE ~



Bookstore trip



Saturday storytime!



Reading at the café



Readability Studio 2026

Readability Test Reference

Blake Madden

Eclipse Foundation AISBL

Brussels

Ottawa

Darmstadt

Readability Studio 2026

Readability Test Reference

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A desk setup featuring a yellow mug with a grid pattern, a blue owl-shaped mug, an open book with a yellow highlighter, and a stack of books including 'Programming Windows' by Charles Petzold.

Preface

This book is a guide to the readability formulas available in the software product *Readability Studio*. Chapters 2, 3, and 4 provide overviews for English, Spanish, and German tests, respectively.



1. Technical Overviews

When writing for a target audience or classifying documents for specific age groups, readability formulas are invaluable tools. They help determine the minimum age group that can quickly process reading material by using an analytical approach. This approach uses factors such as sentence length, word difficulty, and word familiarity.

Most readability scores represent the grade level of the youngest reader that can understand a document. They can also determine how easily a broader audience can process it. For example, most magazine and newspaper articles should aim for scores around the 14–16 age range. This is not because most readers of newspapers are in this age group. It is because this type of material should be easy to read for most age groups. If an article has numerous complex and uncommon words, then it will take much longer to read and understand. (It will also make it less enjoyable to read.) These are not qualities appropriate for something meant to be read quickly by most readers.

Note that these tests only determine the reading level of material based on how easy it is to process. They do not consider the content or themes of a given work. The age level reported by a test score only reflects the age group that can comprehend a document. It does not reflect whether the topics covered in the material are meant for that age group. If you are classifying books into age groups, you should first review them for any mature topics. Then you should use readability tests to determine their comprehension levels.

1.1 How Tests are Calculated

All readability formulas rely on factors such as sentence length, syllable counts, character counts, or word familiarity. Which factors are used vary from test to test, but most tests use between one to three of these factors to gauge a document's difficulty.

Where readability tests may differ is how they are calculated. Most tests are performed using the following methods:

- linear regression equations
- lookup tables
- graphs

The vast majority of tests use multiple linear regressions. Basically, multiple linear regression is a statistical method for modelling the relationship between a dependent variable and two or more independent variables (i.e., predictors). In the case of readability tests, the dependent variable is human-judged grade levels for a set of samples, while the independents are factors such as the samples' sentence lengths and syllable counts. A multiple regression is performed to determine the strongest predictors (usually the top two), and the prediction equation created by the analysis will be the new readability formula.

Another method is lookup tables, where the intersection of factors (e.g., number of sentences and number of syllables) is found in a table. When located, the position in the table will indicate the grade level. Examples of table-based tests include Wheeler-Smith and New Dale-Chall. (Refer to tbl. 2.29 for an example.)

The final method also relies on finding the intersection of factors, but on a graph instead of a table. Most of these plots are divided into bands or brackets, with each of these regions representing a grade or difficulty level. For example, Fry and Raygor use polygonal bands, while Flesch Reading Ease uses brackets along a "ruler":

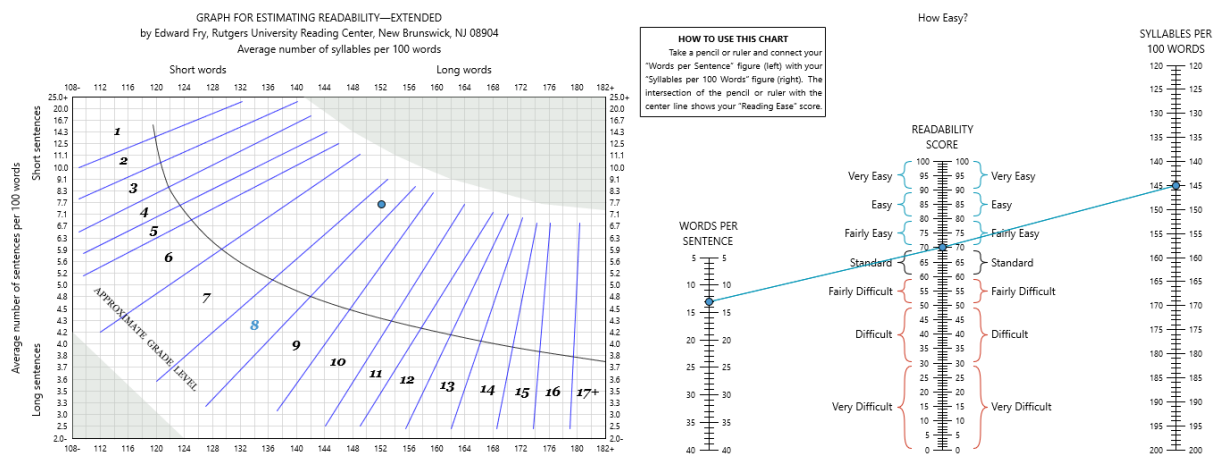


Figure 1.1: Comparison of bands vs. brackets

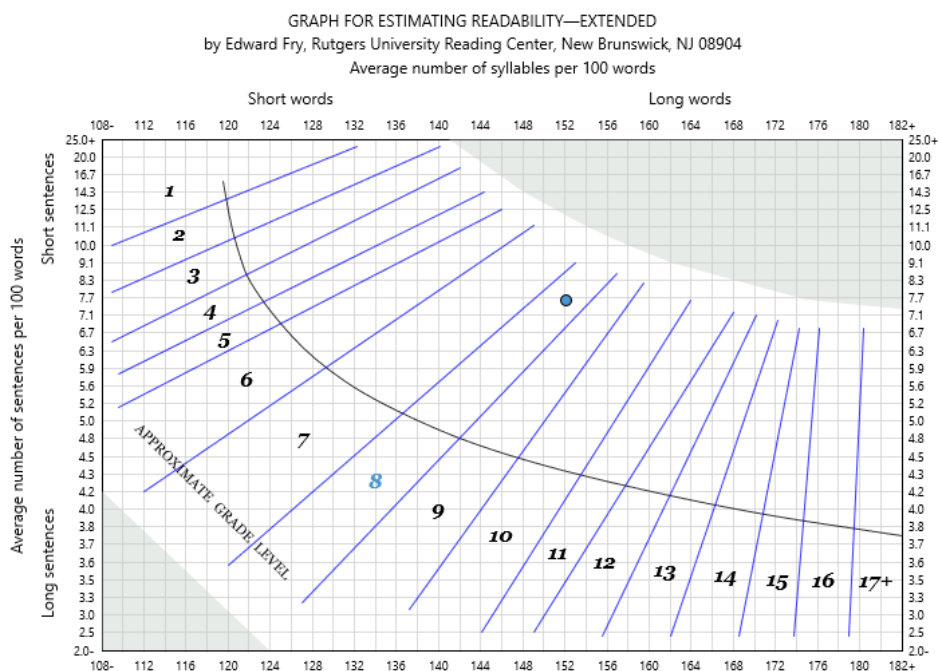
Once the intersection of two factors is located on the plot, the band or bracket that the intersection falls into will indicate the score.

As a final note, there are a few hybrid tests that can be calculated via a regression equation or graph. These tests were initially constructed as regression equations, and this is the preferred method for

performing them. The tests' authors, however, would also offer a graphical version as a quicker and simpler way for performing these tests. Examples of these tests include Flesch Reading Ease and Crawford.

1.2 Grade-level Clamping

A key difference between graph and table-based tests versus regression-based tests is how the former has defined upper limits for their scores. For example, graph-based tests acquire their scores by finding the intersection of certain factors (e.g., sentence length) within a set of bands. These bands represent grades, and there are a specific set of grade bands available. As an example, a Fry graph has bands ranging from 1st grade to 17th (i.e., college):



Likewise, table-based tests take a set of factors and look up the score from a table. These tables have a specific set of grade levels and will also have an upper limit. (Refer to tbl. 2.29 for an example.)

In contrast, tests that use regression equations do not have upper limits. In theory, a document with extreme factors (e.g., abnormally long sentences) could yield scores above grade 19 (i.e., PhD level). To keep results sensible, such scores should be clamped to 19 and be displayed as 19+. This is done for a number of reasons.

First, scores such as 27th grade do not reflect reality. To describe a document as only being understandable by a PhD is relatable. To say that a document can only be processed by someone in the 27th grade, however, makes little sense. Another way to view this is that 19+ implies a document's failure to be readable—similar to an *F* on a test.

Second, because graph and table-based tests are bound to sensical grade scales, it is reasonable to treat regression-based tests the same way. For example, a document scoring 16+ for New Dale-Chall should clamp tests such as Flesch-Kincaid to 19+, thus keeping their scores on a similar scale. In other words, grade and table-based tests were designed for grade scales from 0–19 (sometimes

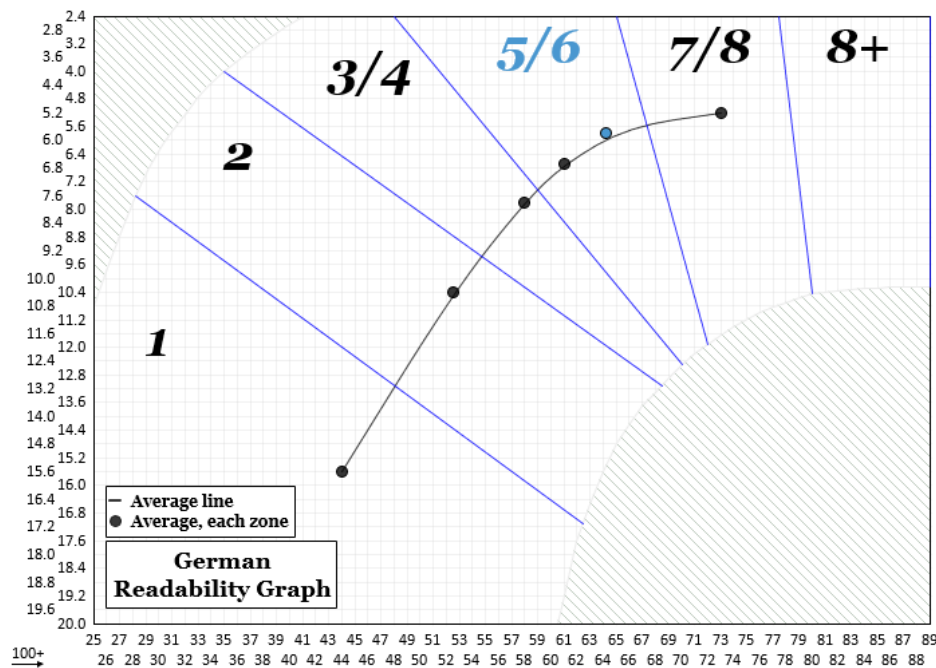
even lower). Likewise, it is reasonable to assume that was the intention of regression-based tests as well.

Finally, the source materials for these formulas cite example scores within 0–19. Again, it can be inferred that the authors' intention was for documents to be bound to a reasonable grade scale.

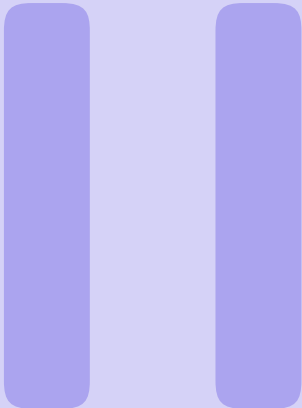
1.3 How Tests Can Fail

Most readability tests will return a result, even when the document being reviewed contains extreme difficulty factors. For example, with table-based tests, documents with long sentences and/or numerous unfamiliar words will fall within the top bracket of the lookup table. (Refer to tbl. 2.29 for an example). Likewise, plugging a document's statistics into a regression-based test will also produce a result without issue (although extreme results may need to be clamped, see sec. 1.2).

It is possible, however, for certain graphical tests to not return a meaningful result. Tests such as Fry and Schwartz use plots that are divided into bands (i.e., polygonal strips) which represent different grade levels. To produce a result, these tests find the intersection of a document's statistics along the plot's axes. Once the intersection is found, the band that the point falls within will indicate the grade level of the document.

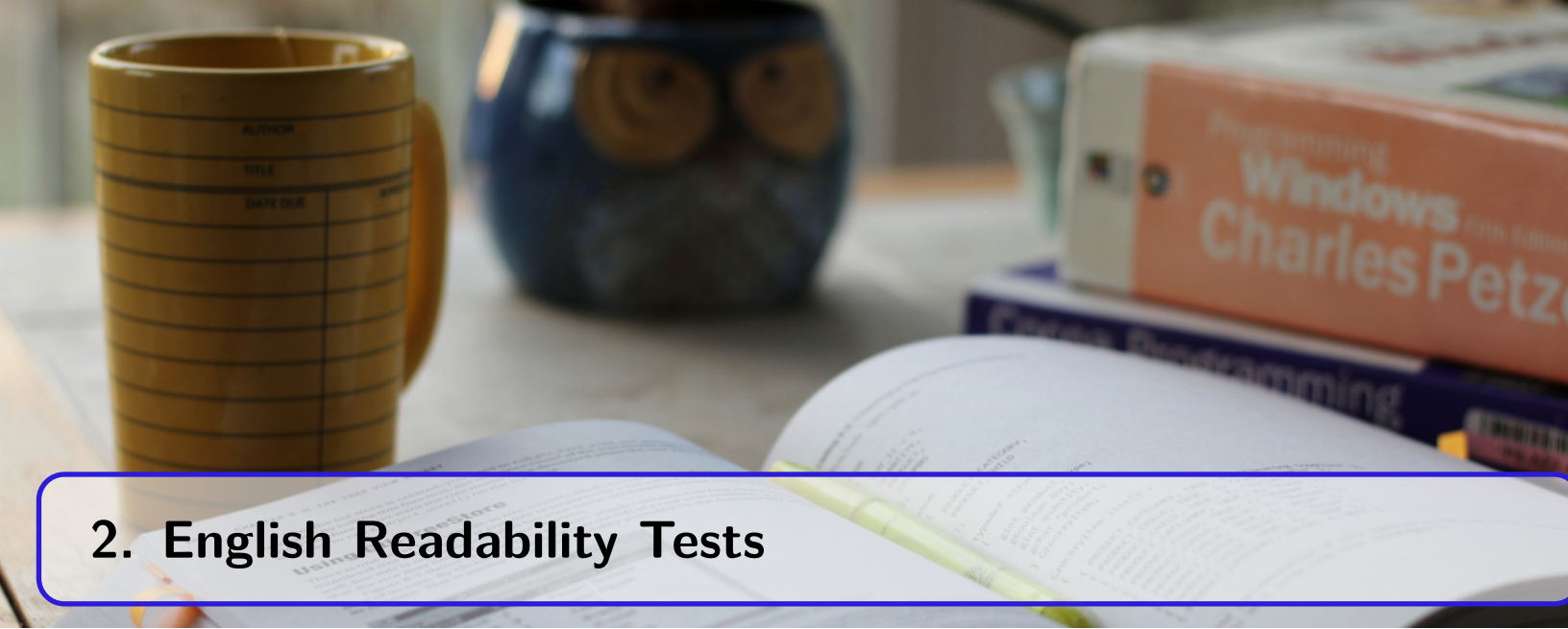


For most documents this will yield a result, but extreme documents may fall outside the grade-level bands. (This can happen if there are too many difficult words or lengthy sentences.)



Readability Tests

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2. English Readability Tests

2.1 Automated Readability Index

The Automated Readability Index—also called “ARI” or “auto”—(E. A. Smith and Senter) was originally created for U.S. Air Force materials. As such, it was designed for technical reports and manuals. This test calculates the grade level of a document based on sentence length and character count.

Kincaid et al. (3, 6–15) also used this test for evaluating peer-prepared reading materials for three studies involving:

- 33 remedial high school students
- 22 graduate students from Southern Georgia College
- 14 trainees participating in a Public Service Careers project at Southern Georgia College

They noted that for the high school study, the materials with lower ARI scores yielded better comprehension rates “as reflected in cloze scores” (3). The graduate students and trainees from the other studies stated that the materials may not be appropriate for remedial high school readers (13, 15), but did agree that the stories were “authentic” and “interesting.”

ARI formula

$$G = (4.71 * (RP/W)) + (0.5 * (W/S)) - 21.43$$

Where:

G	Grade level
W	Number of words
RP	Number of strokes ^a
S	Number of sentences

^a Strokes are characters and punctuation (except for sentence terminating punctuation)

2.2 Bormuth Cloze Mean

The Bormuth Cloze Mean (Machine Computation for Passage Readability) Formula ("Readability: A New Approach" 79–132) calculates a predicted cloze score for a document.

This test is influenced by three variables (word length, sentence length, and word familiarity). Because it uses both word familiarity and syntactic complexity for its analysis, it is often recommended for student documents.

For this formula, a study involving 2,600 students ranging from 4th–12th grade (Development of Readability Analyses 97) was conducted. The students completed 330 110-word cloze passages, and their scores were correlated against numerous linguistic factors from the passages. After a series of factor analyses and multiple regressions, a series of formulas were created to predict cloze and grade-level scores for a passage. These formulas were then cross-validated against 20 250–300 word passages (105). These formulas correlated with between $r^2 = 0.81$ to 0.83 with the original test data and between $r^2 = 0.92$ to 0.93 with the cross validation data.

Of these formulas, the predicted cloze formula is as follows:

Bormuth Cloze Mean formula

$$C = .886593 - .083640 * (R/W) + .161911 * (D/W)^3 - .021401 * (W/S) + .000577 * (W/S)^2 - .000005 * (W/S)^3$$

Where:

C	Estimated cloze score
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

Note

Cloze scores returned by this test should be in percentage format (e.g., 75) rather than fractional format (e.g., .75) to be consistent with other tests.

2.3 Bormuth Grade Placement (GP35)

The Bormuth Grade Placement (Machine Computation for Passage Readability) Formula ("Readability: A New Approach" 79–132) calculates a grade level for a document.

This test is influenced by three variables (word length, sentence length, and word familiarity). Because it uses both word familiarity and syntactic complexity for its analysis, it is often recommended for student documents.

For this formula, a study involving 2,600 students ranging from 4th-12th grade (Development of Readability Analyses 97) was conducted. The students completed 330 110-word cloze passages, and their scores were correlated against numerous linguistic factors from the passages. After a series of factor analyses and multiple regressions, a series of formulas were created to predict cloze and grade-level scores for a passage. These formulas were then cross-validated against 20 250–300 word passages (105). These formulas correlated with between $r^2 = 0.81$ to 0.83 with the original test data and between $r^2 = 0.92$ to 0.93 with the cross validation data.

Of these formulas, the grade-placement formula is as follows:

Bormuth Grade Placement formula

$$G = 3.761864 + 1.053153 * (R/W) - 2.138595 * (D/W)^3 + .152832 * (W/S) - .002077 * (W/S)^2$$

Where:

G	Grade level
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

Note that this particular grade-placement test uses the regression equation that best fit the passages with cloze scores averaging ~35%. There are also 45% and 55% cloze-score fitting Bormuth equations, but as Bormuth noted "35 per cent on a cloze readability test tentatively seemed to represent a criterion for deciding whether or not students are able to exhibit a maximum of information gain as a consequence of reading the passage" (102).

2.4 Coleman-Liau

The Coleman-Liau formula calculates the grade level and estimated cloze score of a document based on sentence length and character count.

Coleman and Liao noted the difficulties of using computers to count syllables (283) for readability formulas. As a solution, they presented a new formula that used character counts as an alternative to syllable counting. This formula was derived from a regression equation trained on the 36 150-word passages from the Miller and Coleman cloze study (851). During validation, the factors of sentences and characters per 100 words yielded a strong correlation of $r^2 = .92$ against Miller and Coleman's cloze scores.

Coleman-Liau calculates the predicted cloze percentage with the following formula:

Coleman-Liau formula

$$E = (141.8401 - (.214590 * R)) + (1.079812 * S)$$

Then convert the cloze percentage into a grade level with the following formula:

$$G = (-27.4004 * (E/100)) + 23.06395$$

Where:

E	Estimated cloze %
G	Grade level
R	Number of characters (per 100 words)
S	Number of sentences (per 100 words)

Though not explicitly mentioned in their article (M. Coleman and Liao 284), the estimated cloze % from the first equation must be converted to a fraction (i.e., divided by 100) to produce the expected results. For this reason, " $(E/100)$ " is shown in the second equation, rather than just "estimated cloze %" as it appears in the source article. As an example, an estimated cloze % of 51.3 should yield a grade level of 9:

$$9 = (-27.4004 * (51.3/100)) + 23.06395$$

2.5 Danielson-Bryan 1

Danielson-Bryan 1 (206) is meant for student materials and calculates the grade level of a document based on sentence length and character count.

DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. This performance benefit is more of a historical reference now; modern computers can calculate syllable counts optimally.

This test was trained against 383 selections from the 1950 edition of McCall-Crabbs *Standard Test Lessons in Reading*. For validation, the derived formula had an $r^2 = 0.575$, similar to Farr-Jenkins-Paterson's $r^2 = 0.553$. This regression equation is as follows:

Danielson-Bryan 1 formula

$$G = 1.0364 * (RP/W) + .0194 * (RP/S) - .6059$$

Where:

G	Grade level
W	Number of words (see below)
RP	Number of characters ^a
S	Number of sentences

^a In this context, characters are letters, numbers, and punctuation (except for sentence-ending punctuation)

In the original article, the statistic of “number of spaces between words” is suggested, rather than explicitly saying “number of words.” Logically, this should be the number of words minus 1. However, in the article’s example, number of spaces between words is the same as number of words (even when there are dashes connecting words). It is therefore assumed that the authors’ intention was to count the number of words using the following logic:

- Count the number of spaces.
- Treat dashes connecting words as spaces.
- Add 1, considering how there would be one less space than words.

For this reason, the *number of words* statistic is a more accurate description of what the authors intended.

2.6 Danielson-Bryan 2

Danielson-Bryan 2 (206) calculates an index score of a document based on sentence length and character/punctuation count. It is a variant of Flesch Reading Ease, with the difference being that it uses character and punctuation counts instead of syllable counts. Being a variant of Flesch, its scores range from 0–100 (the higher the score, the easier to read), with average documents being within 60–70.

DB2 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. Rather than counting syllables, DB2 counts characters and punctuation, which was simpler to analyze on computers of that era. This performance benefit is more of a historical reference now; modern computers can calculate syllable counts optimally.

This test was trained against 376 passages from *McCall-Crabbs Standard Test Lessons in Reading* (1926 ed.). For validation, the derived formula had an $r^2 = 0.575$, similar to Farr-Jenkins-Paterson's $r^2 = 0.553$. This regression equation is as follows:

Danielson-Bryan 2 formula

$$I = 131.059 - 10.364 * (RP/W) - .194 * (RP/S)$$

Where:

I	Flesch index score
W	Number of words ^a
RP	Number of characters ^b
S	Number of sentences

^a Refer to Danielson-Bryan 1 for an explanation of why number of words is used, instead of number of spaces between words.

^b In this context, characters are letters, numbers, and punctuation (except for sentence-ending punctuation)

Table 2.4: Danielson-Bryan 2 (Flesch Reading Ease) Index Table

Flesch Score	Description
90–100	Very easy, third-grade level
80–89	Easy, fourth-grade level
70–79	Fairly easy, fifth-grade level
60–69	Standard, sixth-grade level
50–59	Fairly difficult, junior high school level
30–49	Difficult, high school level
0–29	Very difficult, college level

2.7 Degrees of Reading Power (GE)

The Degrees of Reading Power (grade equivalent) (Carver 303–16) test calculates the difficulty level of a document by performing the DRP test and then converting its units score into a grade level.

This test was trained on the same 330 110-word passages used to build the Bormuth test, along with an additional 30 samples. These additional samples were 100-word passages tested by students from the graduate library at the University of Maryland (305–06). Using these samples (and their cloze and Rauding scores), Carver built a table to convert between DRP units and grade levels (see below). Next, he compared the DRP-GE results against DRP units, Rauding scale scores, and cloze scores. The DRP-GE (grade equivalency) results strongly correlated with these other scores:

- DRP: $r = 0.98$
- Rauding Scale-GE: $r = 0.84$
- cloze: $r = -0.81$

To calculate the grade equivalency, first calculate the DRP difficulty of a document. Then, look up the DRP score from the following table to find its respective grade level:

DRP-Difficulty	DRP-GE	DRP-Difficulty	DRP-GE
0–39	1.5	59	6.8
40	1.6	60	7.3
41	1.7	61	7.8
42	1.7	62	8.3
43	1.8	63	8.8
44	2.0	64	9.4
45	2.1	65	10.0
46	2.3	66	10.6
47	2.5	67	11.2
48	2.8	68	11.8
49	3.0	69	12.5
50	3.3	70	13.1
51	3.6	71	13.7
52	3.9	72	14.4
53	4.3	73	15.0
54	4.7	74	15.7
55	5.1	75	16.4
56	5.5	76	17.1
57	5.9	77	17.9
58	6.3	78–100	18.0

2.8 Degrees of Reading Power

The Degrees of Reading Power (Kibby 416–27; Carver 303–16) formula calculates the difficulty level of a document in terms of DRP units. (These units range from 0 [easy] to 100 [difficult]).

The DRP difficulty score of a document is used to match it to a student based on their DRP ability score. DRP ability tests are manual tests administered to students to gauge their reading levels.

This test is based on the Bormuth Cloze test and uses the same criteria for its calculation. Basically, this test uses the Bormuth formula to calculate a predicted cloze score, converts it to percentage format, and then inverts it to arrive at the units score.

Degrees of Reading Power formula

$$U = 100 - 100 * (.886593 - .083640 * (R/W) + .161911 * (D/W)^3 - .021401 * (W/S) + .000577 * (W/S)^2 - .000005 * (W/S)^3)$$

Where:

U	Units score
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

2.9 Easy Listening Formula

The Easy Listening Formula is designed for “listenability” and is meant for radio and television broadcasts. Fang (65) stated that broadcasted news generally has much shorter sentences compared to print media. Therefore, he created the ELF test to focus solely on complex-word density, not using sentence length as a factor. That is, rather than looking at a sentence’s word count, the density of syllables in a sentence is used instead.

This test was trained on 36 television scripts from the following:

- ABC
- CBS
- NBC
- Local newscasts from the Los Angeles network stations KABC-TV, KNXT, and KNBC

To compare listenability to readability, the study was also trained on 36 newspaper samples from the following:

- *The New York Times*
- *The Wall Street Journal*
- *The Christian Science Monitor*
- *The Los Angeles Times*
- *The Chicago Tribune*
- *The St. Louis Post-Dispatch*

The newscasts yielded the following ELF ranges:

- ABC: 9.9–12.0
- CBS: 9.6–11.9
- NBC: 8.7–10.7
- KNXT: 11.5–13.1
- KNBC: 10.2–12.3
- KABC-TV: 6.4–9.0

While the newspaper samples yielded the following:

- *The New York Times*: 16.8–18.2
- *The Wall Street Journal*: 13.8–15.1
- *The Christian Science Monitor*: 10.3–13.1
- *The Los Angeles Times*: 14.7–18.0
- *The Chicago Tribune*: 14.2–18.6
- *The St. Louis Post-Dispatch*: 11.3–17.1

Note the stark difference between the printed and televised-material scores. This indicates a much higher complex-word density in printed news versus televised news.

To further validate the test, Fang compared Flesch Reading Ease scores against ELF and found a strong correlation of $r = 0.96$.

The ELF score of a sentence is calculated by counting the number of syllables above one for each word. For example:



In any sentence, count each syllable above one per word.



The above sentence contains four polysyllabic words: *any* (2), *sentence* (2), *above* (2), and *syllable* (3). In this case, *any*, *sentence*, and *above* each have 1 syllable above 1, and *syllable* has 2 syllables above one. Adding these values will yield a score of 5.

This process is repeated for every sentence, and then these scores are averaged. A simpler method is shown below:

Easy Listening formula

$$G = (B - W) / S$$

Where:

G	Grade level
W	Number of words
B	Number of syllables
S	Number of sentences

2.10 McAlpine EFLAW

The McAlpine EFLAW[®] formula (McAlpine) is used to determine the ease of reading English text for ESL/EFL (English as a Second/Foreign Language) readers. This test calculates the index score of a document based on sentence length and number of miniwords (i.e., words that contain three or less letters).

Where most other formulas penalize longer words, this test does the opposite, penalizing the use of miniwords. This is because it is designed for ESL readers, as McAlpine explains: "Miniwords are confusing because they have many meanings and are often a sign of wordiness or idioms. . . . these flaws bamboozle EFL readers."

To improve a score, McAlpine recommends lowering the use of miniwords and shortening all sentences to 20 words or less. She also recommends avoiding the following:

- double negatives
- negative questions (e.g., "Are you *not* currently an Ohio resident?")
- idioms/clichés (e.g., "tip of the iceberg")
- helper (auxiliary) verbs (e.g., "can," "may")
- abstract nouns (e.g., "peace," "tension")
- use of pronouns that may be unclear (e.g., "Gabi visited her mother after *she* was released from the hospital.")

McAlpine EFLAW formula

$$I = (W + T)/S$$

Where:

I	The index score
W	Number of words
T	Number of miniwords
S	Number of sentences

Table 2.8: McAlpine EFLAW Index Table

EFLAW Score	Description
1–20	Very easy to understand
21–25	Quite easy to understand
26–29	Mildly difficult
30+	Very confusing

2.11 Farr-Jenkins-Paterson

The Farr-Jenkins-Paterson readability formula (333–37) calculates the Flesch difficulty level of a document based on sentence length and number of monosyllabic words.

Farr-Jenkins-Paterson was designed as a simpler way to calculate a Flesch score because it uses monosyllabic word counting instead of counting all syllables.

This test was trained on 360 100-word samples, producing the following equation:

Farr-Jenkins-Paterson formula

$$I = -31.517 - (1.015 * (W/S)) + (1.599 * ((M/W) * 100))$$

Where:

I	Flesch index score
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

For validation, Flesch and FJP scores from the aforementioned samples were compared, with a significant correlation of $r = 0.93$.

2.12 Flesch-Kincaid

The Flesch-Kincaid readability formula is Kincaid et al.'s modified version of Flesch Reading Ease (Derivation 14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. In addition to being a recalculation, this variation of Flesch returns a grade-level result, rather than an index score. This test calculates the grade level of a document based on sentence length and syllable count.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

Flesch-Kincaid formula

$$G = (11.8 * (B/W)) + (.39 * (W/S)) - 15.59$$

A simplified variation was also provided:

$$G = (12 * (B/W)) + (.4 * (W/S)) - 16$$

Where:

G	Grade level
W	Number of words
B	Number of syllables
S	Number of sentences

Initially, this test was named New Reading Ease Formula (14), but was later referred to as Flesch-Kincaid (CRES 19).

Note that this test sounds out numerals' digits when syllabizing. For example, "1918" will be counted as four syllables ("one"- "nine"- "one"- "eight").

Note

The simplified version of this formula uses a lower precision calculation. It was originally offered for making a manual calculation of this test easier. With the modern availability of computers, it is recommended to use the higher-precision formula.

2.13 Flesch-Kincaid (Simplified)

Refer to sec. 2.12.

2.14 Flesch Reading Ease

The Flesch Reading Ease readability formula calculates an index score of a document based on sentence length and number of syllables.

Flesch Reading Ease (*Art of Readable Writing* 213; *Plain English* 23) is best suited for school textbooks and technical manuals. It is a standard used by many U.S. government agencies, including the U.S. Department of Defense. Scores range from 0–100 (the higher the score, the easier to read) and average documents should be within the range of 60–70.

Flesch Reading Ease formula

$$I = 206.835 - (84.6 * (B/W)) - (1.015 * (W/S))$$

Where:

I	Flesch index score
W	Number of words
B	Number of syllables
S	Number of sentences

Table 2.12: Flesch Reading Ease Table

Flesch Score	Description
90–100	Very easy
80–89	Easy
70–79	Fairly easy
60–69	Standard (plain English)
50–59	Fairly difficult
30–49	Difficult
0–29	Very difficult

Note

Note that this test treats numerals as monosyllabic words by default.

Along with the raw score, a chart is also available to visualize the score's meaning:

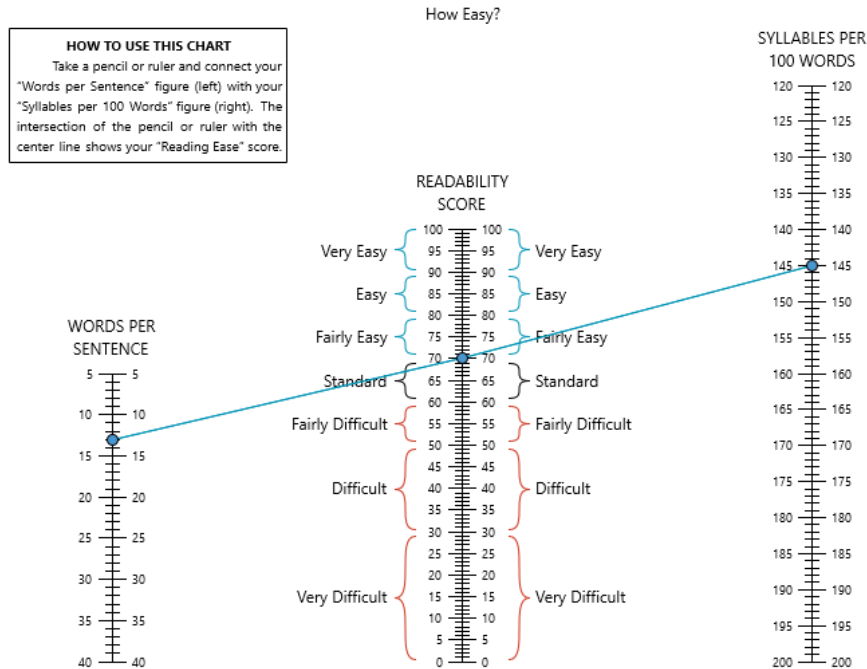


Figure 2.1: Flesch chart

This chart consists of three "rulers." The middle ruler displays the Flesch score as a red point. On both sides of this ruler are brackets showing the reading levels. The bracket that the point falls into will indicate the reading level of the document. The rulers on the left and right of the score ruler will display the factors used to calculate the score. A straight line is drawn between these points to demonstrate how the factors calculated the score.

In the above chart, we can see that the document's average words per sentence is 13 and its syllables per 100 words is 146. This yields a score of 71, which indicates that the document is "fairly easy" to read.

2.15 FORCAST

The FORCAST readability formula (Caylor et al. 13–17) was devised for assessing U.S. army technical manuals and forms. It is the only test not designed for running narrative, so it is best suited for multiple-choice quizzes, applications, entrance forms, etc. This test calculates the grade level of a document based on its number of monosyllabic words.

This formula was trained against U.S. Army “job reading material” (13), using the criterion of cloze passages where readers scored around 35% comprehension. (Most other tests usually use the 50% criterion.) The researchers initially tested regression equations using 15 variables for their new formula, but noted the single factor of monosyllabic words yielded a “sufficiently high” (15) correlation of $r = 0.86$ against the mean cloze scores. Based on this, the following formula was derived:

FORCAST formula

$$G = 20 - (M/10)$$

Where:

G	Grade level
M	Number of monosyllabic words (per 150-words)

A follow-up study for U.S. Air Force regulation materials was later administered (Hooke et al. 13–21). This study included 900 Air Force personnel across 13 bases and used the 40% cloze criterion. The researchers noted during the study that reviewers’ estimated grade levels for regulation documents were generally higher than actual FORCAST scores. Based on this, the authors recommended more extensive use of this test for Air Force materials.

Note that FORCAST results may be slightly different from other tests because it does not take sentence length into account. If your document is structured mostly with tables and lists, then expect some variance between the FORCAST grade level and other tests’ grade levels.

This test is designed for a 150-word sample, but normalization can be used to analyze entire documents.

Note

The name FORCAST is a pseudo acronym for its authors: J. Patrick FORd, John S. CAylor, and Thomas G. STicht.

2.16 Fry Graph

Fry ("Saves Time" 513–16, 575–78; "Clarifications, Validity, and Extension" 242–52) is a graphical test for English documents. It calculates a document's grade level from its average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

The original version of this graph included grade levels 1–12 and college ("Saves Time" 513–516; *Elementary Reading Instruction* 217). It was trained against samples from the following books:

- *Light in the Forest*
- *Of Mice and Men*
- *The Pearl*
- *Shane*
- *Death be Not Proud*
- *The Moon is Down*
- *To Kill a Mockingbird*
- *A Tale of Two Cities*
- *Silas Marner*
- *Act One*

The graph's results were compared to other tests, yielding the following correlations:

- Dale-Chall: $r = 0.94$
- Flesch Reading Ease: $r = 0.96$
- Botel: $r = 0.78$
- SRA: $r = 0.98$

In 1977, Fry extended the graph to include grades 13–17 by extrapolating the averages from the preceding 3 years on the graph ("Clarifications, Validity, and Extension" 251).

The Fry graph is designed for both primary and secondary-age reading materials. Below is an example of a Fry graph:

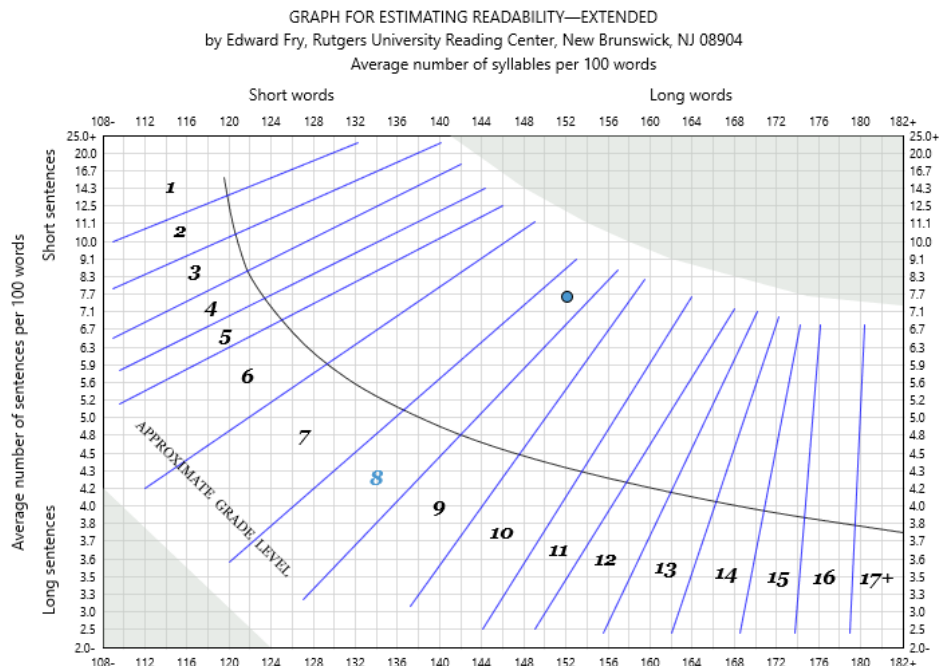


Figure 2.2: Fry graph

Note the following aspects of this graph:

- Grade levels from 1 to 17+ are represented by bands along the middle. If a score falls within one of these bands, then that will be the document's grade level.
- A curved line running through the middle of the grade-level bands. This represents the "smoothed mean of the sample passages. If you plot a large number of passages with a wide range, they will tend to fall somewhere near the line" (243).
- Shaded areas in the bottom-left and top-right corners. These represent the long-sentences and long-words regions, respectively. If a document's score falls into either of these regions, then one of these factors is making the document too difficult to fit into a grade level.

The Fry graph instructions are:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Count the number of syllables in each passage.
- Find the point on the chart.
- If the sample's syllable or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 187 syllables per 100 words, then this will need to be adjusted to 182.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

Note that numerals are fully syllabized for this test. For example, 1945 will be counted as four syllables (*one-nine-four-five*).

2.17 Gunning Fog

The Gunning Fog readability formula (39–41) calculates the grade level of a document based on its number of sentences and complex words.

Complex words are words that contain three or more syllables, with the exception of:

- Proper nouns.
- Words made three syllables by adding *ed* or *es* (e.g., *created* or *trespasses*).
- Compound words composed of simpler words. (Note that forward slashes (/) are treated like hyphens in this context.)

This test can optionally count independent clauses as separate sentences, as *The Technique of Clear Writing* instructs. It should be noted, however, that this can cause misleadingly low scores for documents that contain run-on sentences. Consider the following:

// The Meldorf® statistical library is available in C++ source code—a low-level programming language—and in the form of a COM library—a Microsoft® technology for inter-program communication; the COM library can be called from numerous languages, such as: Visual Basic, C#, and Delphi. //

If you treat independent clauses as separate sentences, then this will be seen as six small sentences and produce a Fog score of 10.2. If you count sentences by only looking for ., ?, or !, then this will be counted as a single 44-word sentence and produce a Fog score of 19. To penalize run-on sentences (which is recommended), then leave the option Count independent clauses as separate sentences unchecked for this test.

Gunning Fog formula

$$G = .4 * (W/S + ((F/W) * 100))$$

Where:

G	Grade level
W	Number of words
F	Number of complex words
S	Number of sentences

2.18 Harris-Jacobson Wide Range

The Harris-Jacobson Wide Range Readability Formula test (19–30) calculates the grade level of a document based on sentence length and number of unique unfamiliar words. Words are unfamiliar if they do not appear on a list of common words that are known to most second-grade students. The following are also considered familiar:

- Regular verb forms of any common word (i.e., *ing*, *es*, *s*, *ed*, and *ies* endings). Irregular forms are not familiar.
- Regular plural and possessive forms of any common word (i.e., *s*, *es*, and *ies* endings). Irregular forms are not familiar.
- Adjectival or adverbial forms of common words. This includes forms with endings such as *ly*, *est*, *er*, and *ily*.
- Hyphenated words if both parts of the word are common words. For example, *apple-tree* and *and/or* would be familiar. (Note that forward slashes (/) are treated like hyphens in this context.) Although this rule is not explicitly stated in *Basic Reading Vocabularies*, it is recommended because both Spache and New Dale-Chall apply it.
- All proper nouns.

Note that numerals are entirely excluded from this test and are considered neither familiar nor unfamiliar.

The Wide Range Readability Formula is based on previous Harris-Jacobson formulas designed for preprimer through eighth-grade reader-level materials. The Wide Range variation was trained on a new series of basal readers (30). This series included:

Primary Levels	Secondary Levels
1 preprimer	1 fourth reader
1 primer	1 fifth reader
1 first reader	1 sixth reader
2 second readers	1 seventh reader
2 third readers	1 eighth reader

Using these new basal readers as the test's criterion, a new regression equation and expanded word list were built for the Wide Range formula. This new formula could significantly correlate samples from these books to their reported reading levels. Further, to predict readability scores for materials beyond the eighth-grade level, the lookup table used by the test (see below) was extrapolated to include secondary and adult-level materials.

Note

This test treats headers and subheaders as full sentences, but excludes lists and tables. The default text exclusion method will be overridden for this specific test.

Harris-Jacobson is calculated from the following steps:

- Obtain the V1 score: (Number of unique unfamiliar words / Number of words) * 100.
- Obtain the V2 score: Number of words / Number of sentences.
- Multiply V1 by .245.
- Multiply the V2 score by .160.
- Add together the result of step 3, step 4, and .642 to achieve the predicted raw score.
- Round off the predicted raw score to one decimal place.
- Find the predicted raw score in the following table to acquire the final readability score:

Predicted Raw Score	Readability Score	Predicted Raw Score	Readability Score
1.1	1.0	4.6	4.8
1.2	1.0	4.7	5.0
1.3	1.0	4.8	5.2
1.4	1.1	4.9	5.4
1.5	1.2	5.0	5.5
1.6	1.3	5.1	5.7
1.7	1.4	5.2	5.9
1.8	1.5	5.3	6.0
1.9	1.7	5.4	6.2
2.0	1.8	5.5	6.4
2.1	1.9	5.6	6.5
2.2	2.0	5.7	6.7
2.3	2.1	5.8	6.9
2.4	2.2	5.9	7.1
2.5	2.3	6.0	7.3
2.6	2.4	6.1	7.5
2.7	2.6	6.2	7.7
2.8	2.7	6.3	7.9
2.9	2.8	6.4	8.1
3.0	2.9	6.5	8.3
3.1	3.1	6.6	8.5 ^a
3.2	3.2	6.7	8.7
3.3	3.3	6.8	8.9
3.4	3.4	6.9	9.1
3.5	3.5	7.0	9.2
3.6	3.6	7.1	9.4
3.7	3.7	7.2	9.6
3.8	3.8	7.3	9.8
3.9	3.9	7.4	10.1
4.0	4.0	7.5	10.3
4.1	4.1	7.6	10.5
4.2	4.3	7.7	10.7
4.3	4.5	7.8	10.9
4.4	4.6	7.9	11.1
4.5	4.7	8.0	11.3

^a Readability scores above 8.5 were derived by extrapolation (21)

Because this formula is based on the usage of familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

2.19 Lix

The Läsbarhetsindex (Lix) readability formula (Björnsson 480–97) was designed for documents of any Western European language. It calculates a document's index score based on sentence length and number of long words (i.e., words containing seven or more characters). Viewing long words as those with seven or more characters is what differentiates it from similar tests (which use six or more characters). This difference is what allows it to be a more balanced “yardstick” that can be applied to numerous languages.

The philosophy behind using one formula for multiple languages is that, at their core, most languages are similar in difficulty. This approach argues that a language having various idiosyncrasies and longer words is not why it scores at more difficult levels. The reason languages score higher is because of how they are used (i.e., word choice, writing style) and presented (e.g., longer paragraphs and sentences, fewer images). This differs from the reasoning behind other tests based on adjusted English formulas. Those tests are designed under the assumption that non-English languages are inherently more difficult (because of comparatively longer words). For example, Gilliam-Peña-Mountain and the German reinterpretation of Lix follow this logic of adjusting English formulas for non-English text. Lix, however, chooses to use the same formula to gauge multiple languages. It views documents that score higher to be a result of writing style and presentation, not the language itself.

This test was originally designed for Danish, English, French, German, and Finnish and trained on “thousands of books and reading passages” (481) during 1968, 1969, 1974, 1975, and 1979. During the 1974 English study, the formula was tested with:

- 20 children's books
- 30 fiction books
- 30 non-fiction books
- 20 technical books

Other language studies were conducted later using newspaper articles as their training material. These languages included Danish, English, Finnish, French, German, Italian, Norwegian, Portuguese, Russian, Spanish, and Swedish. (Anderson conducted a separate study with Greek material in 1981.) The researchers' findings noted the following average scores across the languages:

- Swedish (47), Norwegian (48), and Danish (51) were the lowest scoring (i.e., easiest to read)
- ...followed by English (52), French (55), and German (59)
- ...then Italian (65), Spanish (67), and Portuguese (70)
- ...and Finnish (72) and Russian (65) being the most difficult

The researchers conjectured these differences had more to do with use of language than with the languages themselves (495).

The index score of Lix can be summarized as follows:

Table 2.15: Lix Index Table

Lix Score	Description
0–29	Very easy (books for children)
30–39	Easy (fiction)
40–49	Average (factual prose)

Lix Score	Description
50–59	Difficult (technical literature)
60 and above	Very difficult

Note that Lix also includes a chart to visualize the meaning of its index score:

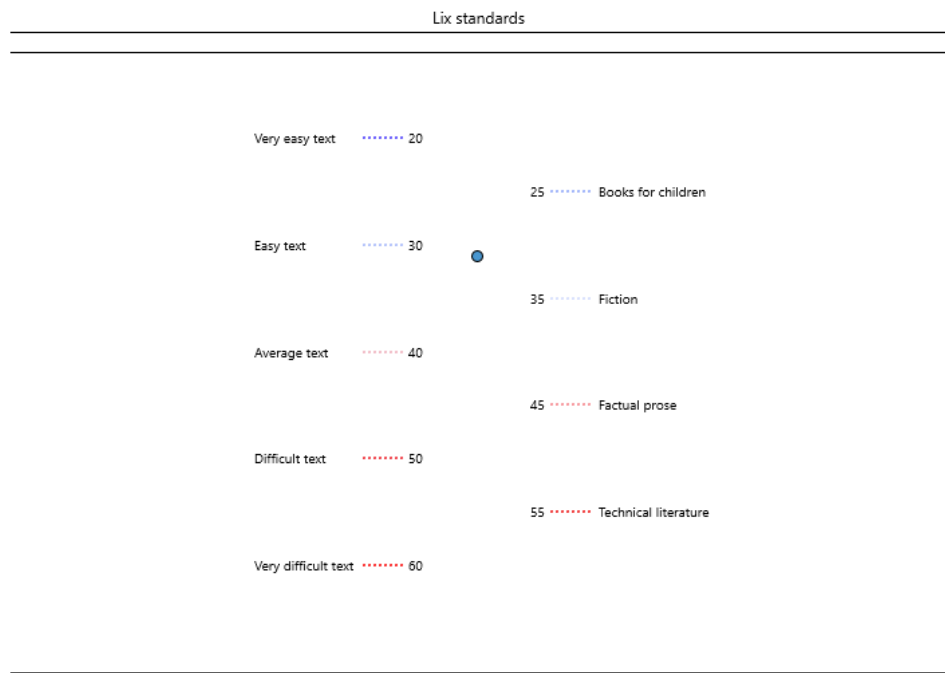


Figure 2.3: Lix gauge

Lix formula

$$I = W / S + 100 * (X / W)$$

Where:

I	Lix index score
W	Number of words
X	Number of long words (7+ characters)
S	Number of sentences

As a final note, Anderson contributed a grade-level conversion table for Lix index scores (Anderson, "Lix and Rix: Variations of a Little-Known Readability Index" 494) as follows:

Table 2.17: Lix Grade-level Conversion Table

Lix Score	Equivalent Grade Level
56+	College
52-55	12
48-51	11
44-47	10
40-43	9
36-39	8
32-35	7
28-31	6
24-27	5
20-23	4
15-19	3
10-14	2
Below 10	1

2.20 Modified SMOG

The Modified SMOG test (L. L. Smith 129–32) calculates the grade level of a document based on sentence length and number of difficult (i.e., 3 or more syllable) words and is designed for primary and secondary-age materials.

Modified SMOG differs from SMOG in the following ways:

- It uses the unique number of difficult words (rather than the total).
- It uses a smaller constant in its formula.
- The score is rounded, rather than floored.

Because of these adjustments, it will generally produce scores closer to other tests than SMOG does—in particular, the Fry graph (129).

The author's decision not to count repeated difficult words was a major factor in making this test more appropriate for primary readers. As he explains, in a particular pre-primer book "the word 'kangaroo' appeared 52 times . . . it was the only polysyllabic word in the book. To count the word more than once would have inflated the readability score to an unrealistic level" (130).

The steps for calculating Modified SMOG are:

- Take 3 ten-sentence samples.
- Count the number of unique words containing 3 or more syllables.
- Calculate the square root of this count and round it.
- If this square root is 4 or more, then add 1. Otherwise, the square root should remain the same. This will be the grade level score.

Note that numerals are fully syllabized (i.e., sounded out) for this test.

This test requires a 10-sentence sample, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

2.21 New Automated Readability Index

The New Automated Readability Index is Kincaid et al.'s modified version of ARI (Derivation 14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. This test calculates the grade level of a document based on sentence length and character count.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Automated Readability Index formula

$$G = (5.84 * (RP/W)) + (.37 * (W/S)) - 26.01$$

Where:

G	Grade level
W	Number of words
RP	Number of strokes ^a
S	Number of sentences

^a Strokes are characters and punctuation (except for sentence-terminating punctuation).

A simplified variation was also provided:

$$G = (6 * (RP/W)) + (.4 * (W/S)) - 27.4$$

Note

The simplified version of this formula uses lower precision. It was originally offered to make manual calculations of this test easier. With the modern availability of computers, it is recommended to use the higher-precision formula.

2.22 New Automated Readability Index (Simplified)

Refer to sec. 2.21.

2.23 New Dale-Chall

The New Dale-Chall readability test (Chall and Dale, *Manual for the New Dale-Chall Readability Formula* 1–7) calculates the grade level of a document based on sentence length and number of unfamiliar words. Words are unfamiliar if they do not appear on a list of 3,000 common words that are known to most 4th grade students. The following are also considered familiar:

- Regular plural and possessive forms of common words (i.e., *s*, *es*, and *ies* endings). Irregular forms (e.g., *oxen*) are not familiar.
- Adjectival and verb forms of common words with the following endings: *d*, *ed*, *ied*, *ing*, *s*, *es*, *ies*, *r*, *er*, *est*, *ier*, *iest*.
- Compound and hyphenated words if both parts of the word are common words. For example, *long-haired* and *and/or* would be familiar. Note that forward slashes (/) are also treated as compound-word punctuation.
- The first instance of any proper noun—all subsequent occurrences are considered familiar.
- All numerals.

Note

This test treats headers and subheaders as full sentences, but excludes lists and tables.

The initial version of this test (Dale and Chall) was a multiple regression trained against 376 passages from McCall-Crabbs *Standard Test Lessons in Reading*. After comparing the results of this formula against other tests (such as Flesch and Lorge), it was validated against manually judged reading levels from other materials. These materials included “fifty-passages of health-education materials” (18) with a correlation of $r = 0.92$. Also included were “78 passages on foreign affairs from current-events magazines, government pamphlets, and newspapers” (18) with a correlation of $r = 0.90$.

The New Dale-Chall test replaces this regression equation by looking up the factors’ intersection in a table and returning a grade range (e.g., 4 or 7–8). (This grade range is a notable difference from other tests that use floating-point precision grade levels.) It also includes an updated version of the 3,000 common words.

Because this test is based on the usage of familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

2.24 New Farr-Jenkins-Paterson

The New Farr-Jenkins-Paterson readability formula calculates the grade level of a document based on sentence length and number of monosyllabic words.

New Farr-Jenkins-Paterson is Kincaid et al.'s recalculated version of the FJP test (Derivation 17) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. In addition to being a recalculation, this variation of FJP (i.e., a Flesch-like test) returns a grade-level result, rather than an index score.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Farr-Jenkins-Paterson formula

$$G = 22.05 + (.387 * (W/S)) - (.307 * ((M/W) * 100))$$

Where:

G	Grade level
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

2.25 New Fog Count

The New Fog Count readability formula is Kincaid et al.'s modified version of the Gunning Fog Index (14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. This test calculates the grade level of a document based on sentence length and number of words containing three or more syllables.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Fog Count formula

$$G = (((Z + (3 * F))/S) - 3)/2$$

Where:

G	Grade level
F	Number of complex words
Z	Number of easy words
S	Number of sentences

 Note

This test uses the same easy-word and sentence counting methods as the Gunning Fog test.

2.26 (Powers-Sumner-Kearl) Dale-Chall

The (Powers-Sumner-Kearl) Dale-Chall readability formula (99–105) calculates the grade level of a document based on sentence length and number of unfamiliar Dale-Chall words.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to “modernize the formulas by taking advantage of the more recently administered tests” and to “establish formulas which are derived from identical materials” (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs’s *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from “various publications” (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Dale-Chall formula is as follows:

(PSK) Dale-Chall formula

$$G = 3.2672 + (.0596 * (W/S)) + (.1155 * ((UDC/W) * 100))$$

Where:

G	Grade level
S	Number of sentences
UDC	Number of unfamiliar Dale-Chall words
W	Number of words

Although this is an update of the 1948 Dale-Chall formula, the 1995 New Dale-Chall test is recommended instead. This formula does not apply as much weight to the sentence length factor as New Dale-Chall does and comparatively yields lower results for difficult (i.e., upper-secondary education) documents. Therefore, this test should only be used for primary and lower-secondary educational documents. It is also only recommended if floating-point precision for the results is required, as New Dale-Chall uses grade ranges instead.

2.27 (Powers-Sumner-Kearl) Farr-Jenkins-Paterson

The (Powers-Sumner-Kearl) Farr-Jenkins-Paterson readability formula (99–105) calculates the grade level of a document based on sentence length and number of monosyllabic words.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to “modernize the formulas by taking advantage of the more recently administered tests” and to “establish formulas which are derived from identical materials” (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs’s *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from “various publications” (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Farr-Jenkins-Paterson formula is as follows:

(PSK) Farr-Jenkins-Paterson formula

$$G = 8.4335 + (.0923 * (W/S)) - (.0648 * ((M/W) * 100))$$

Where:

G	Grade level
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

2.28 (Powers-Sumner-Kearl) Flesch

The (Powers-Sumner-Kearl) Flesch readability formula (99–105) calculates the grade level of a document based on sentence length and number of syllables.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to “modernize the formulas by taking advantage of the more recently administered tests” and to “establish formulas which are derived from identical materials” (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs’s *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from “various publications” (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Flesch formula is as follows:

(PSK) Flesch formula

$$G = (ASL * .0778) + (SY * .0455) - 2.2029$$

Where:

G	Grade level
ASL	Average sentence length
SY	Number of syllables per 100 words

Note

Note that this test treats numerals as monosyllabic words, in conjunction with the original Flesch test.

2.29 (Powers-Sumner-Kearl) Gunning Fog

The (Powers-Sumner-Kearl) Gunning Fog readability formula (99–105) calculates the grade level of a document based on sentence length and number of syllables.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to “modernize the formulas by taking advantage of the more recently administered tests” and to “establish formulas which are derived from identical materials” (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs’s *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from “various publications” (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Gunning Fog formula is as follows:

(PSK) Gunning Fog formula

$$G = 3.0680 + (.0877 * (W/S)) + (.0984 * ((F/W) * 100))$$

Where:

G	Grade level
W	Number of words
F	Number of complex words
S	Number of sentences

Note

This test uses the same easy-word and sentence counting methods as the Gunning Fog test.

2.30 Raygor Estimate Graph

The Raygor Estimate graph (259–63) is a graphical test for English documents. It calculates a document's grade level from its average number of sentences and long (6+ character) words per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

Note that this graph is very similar to the Fry graph. The intent of this test was to provide a quicker and simpler way to plot a readability level, assuming that counting characters is easier than syllables. To confirm this, a secondary study (Baldwin and Kaufman 148–53) conducted an experiment involving 155 students from the University of Tulsa. The students performed Fry and ELF tests on 6 100-word passages. Afterwards, an independent *t* test was performed to gauge the difference in time spent manually performing the two readability tests. What the researchers found was a significant difference, where Raygor took an average of 250.26 seconds to complete versus 323.45 seconds for Fry.

In another experiment, Baldwin and Kaufman compared the correlations between Fry and Raygor scores. They collected three 100-word samples from 100 books from the Learning Resource Center at The University of Tulsa. With these books, the “Levels of difficulty in the sample ranged from upper elementary to professional (beyond the college level)” (149). The researchers found a significant Kendall rank order correlation of 0.875 between scores from the two tests.

This graph is primarily used in secondary education to help classify teaching materials and books into their appropriate reading groups. Below is an example of a Raygor graph:

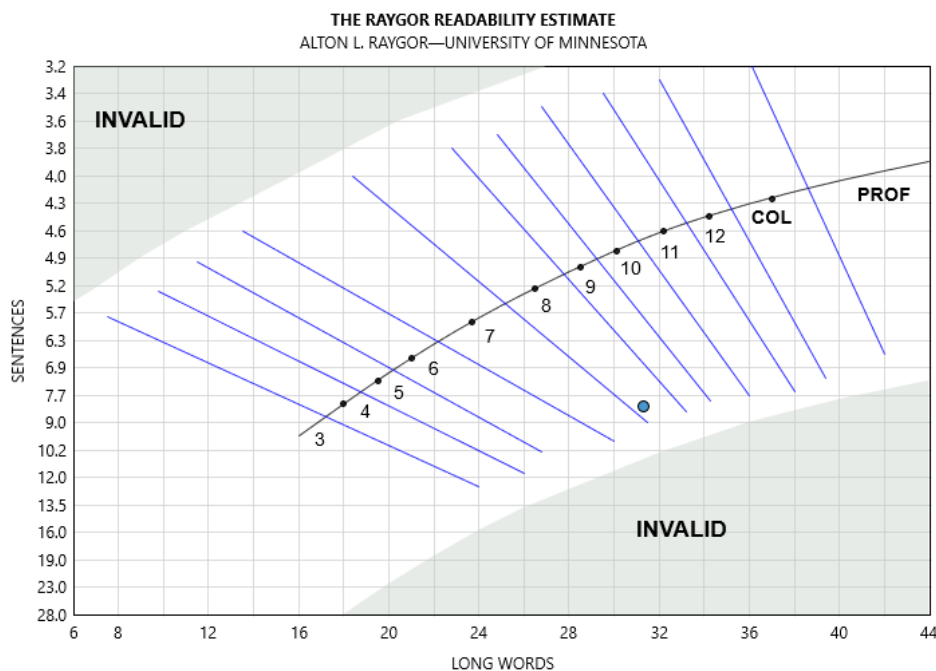


Figure 2.4: Raygor Estimate graph

The Raygor Estimate instructions are:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end. Numerals should be skipped.
- Count the number of sentences, estimated to the nearest tenth.
- Count the number of words that are six or more letters.
- Find the point on the chart.
- If the sample's character or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 30 sentences per 100 words, then this will need to be adjusted to 28.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with three samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

2.31 Rix

The Rate Index (Rix) readability formula ("Lix and Rix" 490–96) was designed for documents of any Western European language. It calculates a document's index score based on sentence length and number of long words (i.e., words containing seven or more characters). It is based on the Lix readability formula (Björnsson 480–97) and follows the same philosophy of using a unified formula for multiple languages.

"Rate" in this context refers to the rate of long words throughout the document. Comparing against other tests that traditionally recommend fixed sample sizes (e.g., SMOG), Anderson noted that Rix could be applied to any length of text without normalization. As he explained, "the concept of rate has the advantage of being applied over any number of sentences, for it is a running average" ("Lix and Rix" 495).

Regarding the factors used for this test, Anderson defined a sentence "as a sequence of words terminated by a full-stop (period), question or exclamation mark, colon, or semicolon" (495). In other words, sentences units should be used for the calculation. (As for Lix, Björnsson did not define how sentences should end, so it is recommended to use the traditional ., ?, and ! markers for that test.)

As for defining long words, Anderson recommended "excluding hyphens, punctuation marks, and brackets" (495) when determining a word's length. Although Björnsson did not explicitly state this for Lix, it is a reasonable assumption that was his original intention.

Prior to creating this formula, Anderson conducted a validation study for Lix, using a set of 36 prose passages. These 150-word passages were originally scored from a cloze study (Miller and E. Coleman 851). Later, they were scored in another study involving judged-rankings and word-recollection tests (Aquino 346–56). (Both portions of Aquino's study used 14 *Southwest Regional Laboratory for Educational Research and Development* employees.) Anderson analyzed the Lix scores against these other measurements and found significant correlations, as shown below:

- against the Miller and Coleman's cloze scores, $r = 0.89$
- against word-recall scores, $r = 0.82$
- against the judges' rankings, $r = 0.88$

After validating Lix, Anderson created Rix and compared the two tests' results. He found that they almost perfectly correlated ($r = 0.99$). Afterwards, he validated Rix against the aforementioned 36-passage studies and found similarly strong correlations.

Rix formula

$$I = X/U$$

Where:

I	Rix index score
X	Number of long words (7+ characters)
U	Number of sentence units

Table 2.25: Rix Grade-level Conversion Table

Rix Index Score	Grade Level
7.2 and above	College
6.2 and above	12
5.3 and above	11
4.5 and above	10
3.7 and above	9
3.0 and above	8
2.4 and above	7
1.8 and above	6
1.3 and above	5
.8 and above	4
.5 and above	3
.2 and above	2
Below .2	1

2.32 SMOG

The SMOG readability formula (McLaughlin 639–46) calculates the grade level of a document based on complex (i.e., 3 or more syllable) word density and is designed for secondary-age readers.

SMOG attempts to predict “full comprehension” (L. L. Smith 129), whereas most other formulas predict 50%–75% comprehension. Because of this, SMOG will generally produce higher scores comparatively (usually 1–2 grade levels higher).

This test was trained on 390 passages from the 1961 edition of *McCall-Crabbs Standard Test Lessons in Reading*. The following regression equation was derived with a correlation of $r = 0.985$:

SMOG formula

$$G = 1.0430 * \sqrt{C} + 3.1291$$

Where:

G	Grade level
C	Number of complex (3+ syllables) words from 30 sentences

A simplified variation is also available (McLaughlin 643):

$$G = FLOOR(\sqrt{C}) + 3$$

FLOOR refers to rounding the result of $\sqrt{C}$ down to the closest perfect square.

Note

The simplified version of this formula uses lower precision. It was originally offered to make manual calculations of this test easier.

This test requires 3 10-sentence samples, but normalization can be used to analyze entire documents.

Numerals are fully syllabized (i.e., sounded out) for this test.

Note that this test is colloquially referred to as an acronym for “Simple Measure of Gobbledygook.” The origin of its name, however, is actually a nod to Robert Gunning’s Fog index.

2.33 SMOG (Simplified)

Refer to sec. 2.32.

2.34 Spache Revised

The Spache Revised test (*Good Reading* 195–207) calculates the grade level of a document based on sentence length and number of unique unfamiliar words. This test uses a list of 1,041 common words that are known to younger readers (4th grade and below). Any word that does not appear on this list is considered unfamiliar. The following are also considered familiar:

- Regular verb forms of any common word (i.e., *ing*, *es*, *s*, *ed*, and *ies* endings). Irregular forms are not familiar.
- Regular plural and possessive forms of any common word (i.e., *s*, *es*, and *ies* endings). Irregular forms are not familiar.
- Adjectival or adverbial forms of common words. This includes forms with endings such as *ly*, *est*, *er*, and *ily*.
- Derivatives of common words where the function changes (e.g., an adjective changed to function as a noun) are not familiar unless explicitly on the common word list. For example, *brave* is a common word, but *bravery* would not be considered familiar.
- Hyphenated words if both parts of the word are common words. For example, *apple-tree* and *and/or* would be familiar. Note that forward slashes (/) are treated like hyphens in this context.
- All proper nouns.
- All numerals.

Spache Revised is generally used for primary-age (i.e., Kindergarten to 7th grade) readers to help classify school textbooks and literature. This differs from New Dale-Chall, which is better meant for secondary age readers.

Because this formula uses familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

Spache Revised formula

$$G = (.121 * (W/S)) + (.082 * UUS) + .659$$

Where:

G	Grade level
S	Number of sentences
W	Number of words
UUS	Number of unique unfamiliar words per 100 words (see note below)

Note

Spache recommends not counting the same unfamiliar word more than once if it appears in subsequent 100-word samples (200). For this reason, it is recommended to follow this practice when analyzing an entire document.

2.35 Stocker's Catholic Supplement

Stocker's Catholic supplement (87–89) is an extension to the Dale-Chall word list designed for Catholic students. Based on a 1971 study of 4th-grade Catholic students¹, this list represents over 200 words (not on the standard Dale-Chall list) familiar to 80% of them.

This list is used along with the Dale-Chall list and offers more accurate scoring with materials meant for Catholic schools.

¹The students involved in this study were "6,743 randomly selected Catholic fourth-graders attending 136 parochial schools throughout the country" (87–88).

2.36 Wheeler-Smith

The Wheeler-Smith (397–99) test is designed for primary-age materials (Kindergarten to 4th grade). This test calculates the grade level of a document based on unit length and number of polysyllabic (2+ syllable) words. It should be noted that this differs from most other tests that use 3+ syllable words as a difficulty factor. The researchers selected 2+ syllable words because primary readers will have more difficulty with them than secondary and adult readers.

Although not explicitly stated in the article, the authors' definition of "polysyllabic" as being 2+ syllable can be inferred from the example² that they provided:

// The Woodsman saw the seven sisters come into their bedroom. He saw them put on the shoes in which they danced. He saw them put on their party dresses. When they were all dressed, the oldest sister opened a door in the floor near her bed. "Sisters, we are ready," she said. They walked down the secret steps. The Woodsman put on his magic cap and followed them down the steps. Thanks to his magic cap, no one was able to see him. Soon they came to a wonderful wood. Some of the trees had blue leaves. Some had glass leaves. Others had leaves of gold. //

The authors reported 20 polysyllabic words in the above sample. *Wonderful* is the only 3+ syllable word in this sample, whereas there are 18 2+ syllable words. It may be inferred that the authors were considering *their* as 2 syllables, which in that case would yield 20 2+ syllable (i.e., "polysyllabic") words.

In regards to unit (i.e., sentence) length, the researchers noted that sentence lengths strongly correlated with the manually-rated grade levels. Because of this, sentence length was given a stronger weight in the formula (compared to previous formulas).

For the building and validation of this formula, nine basic reading series were used.

Finally, the authors offered recommendations while reviewing a reported score. First, if a score appears to be higher than expected, then you should exclude proper nouns and rerun the test. Second, they recommended that when assigning a book for independent reading, assign it one level below the reported score. In other words, if a book scores at the 4th grade level, then it should be assigned to 3rd grade readers if they are reading it independently.

Wheeler-Smith formula

$$I = (W/U) * (P/W) * 10$$

²This sample was taken from *Down Singing River*, Betts-Welch, American Book Company.

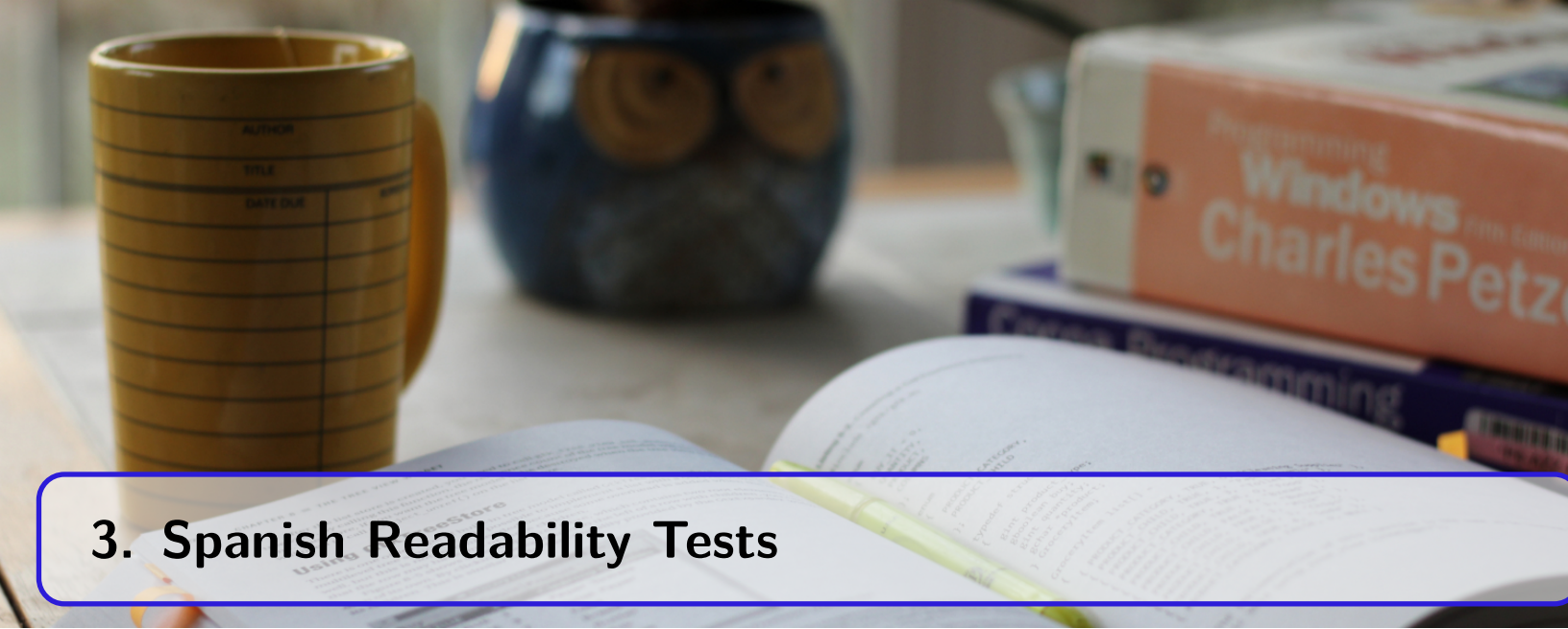
Where:

I	Index value
P	Number of polysyllabic (2+ syllable) words
W	Number of words
U	Number of units

Then, find the index value in the following chart to acquire the final readability score:

Table 2.29: Wheeler-Smith Grade-level Conversion Table

Index Score	Grade Level
4.0–8.0	Primer (Kindergarten)
8.1–11.5	First grade
11.6–19.0	Second grade
19.1–26.5	Third grade
26.6–34.5	Fourth grade



3. Spanish Readability Tests

3.1 Crawford

The Crawford test is designed for “elementary level materials in Spanish” (4). This test calculates the grade level of a document based on sentence length and syllable count.

The test was derived from a multiple regression equation trained against 10 basal readers series collected from the United States, Latin America, and Spain. (These series represented grade levels 1–6 and contained 789 100-word passages in total.) Across these reader series, Crawford noted somewhat significant correlations for syllables per 100 words (an average of $r = 0.4152$) and a consistently stronger correlation for sentences per 100 words (an average of $r = 0.6395$). Using these findings, the following equation was built:

Crawford formula

$$G = (S * -.205) + (B * .049) - 3.407$$

Where:

G	Grade level
S	Number of sentences per 100 words
B	Number of syllables per 100 words

A chart visualizing this test’s score is also available:

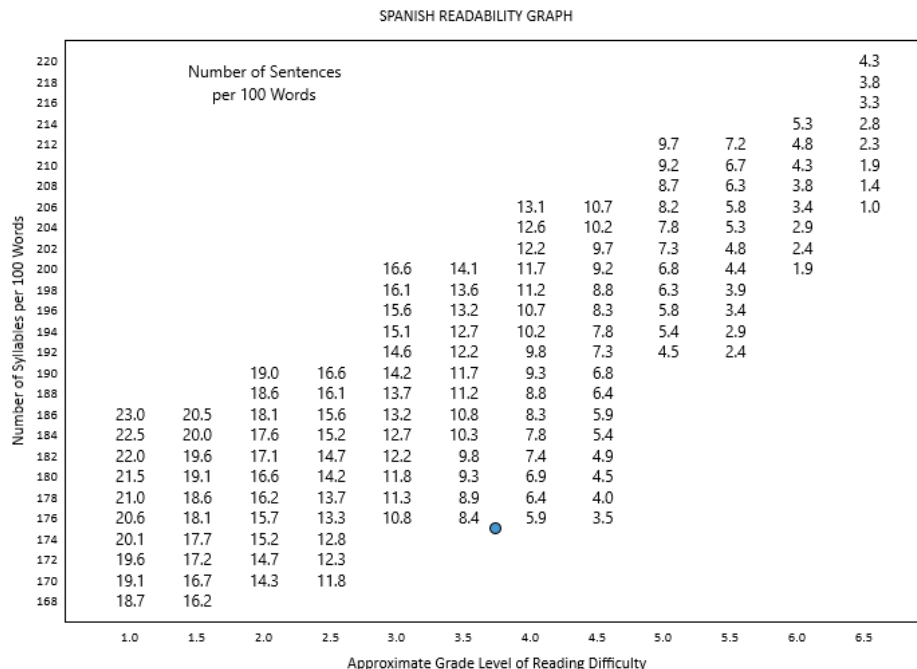


Figure 3.1: Crawford chart

In this chart, the y-axis and interior (i.e., plot area) represent the factors, and the x-axis represents the final score. Specifically, the y-axis represents the sample’s syllable count, and the interior area rep-

resents the sample's sentence count. After finding the intersection of these two factors, the position on the x-axis respective to this intersection yields the grade-level score.

Note that this differs from other graphs which use the x- and y-axes to represent the factors and their intersection to represent the score. Instead, this chart displays the sentence-count values as a series of steps across the plot area.

For example, in the above graph the number of syllables is 175. The number of sentences is approximately 7 (between 8.4 and 5.9 [on the plot, respective to 175 on the y-axis]). After finding the intersection of these factors, the point's x-axis value is 3.7. Hence, 3.7 is the sample's grade-level score.

This test is designed for 100-word samples and uses the same statistics as the Fry test. (Normalization can also be used to analyze larger samples or entire documents.)

3.2 FRASE Graph

FRASE (Fry Readability Adaptation for Spanish Evaluation) (Vari-Cartier 141–48) is a graphical test for Spanish text. This test calculates a document's language level from its average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

The test uses methodologies borrowed from the Fry graph, such as using the same factors (sentence length and syllable count). However, it also includes adjustments which differentiate it. One adjustment is that the x-axis begins at 182 (Fry begins at 108)—this is used to account for Spanish text having higher syllable counts compared to English. The second adjustment is that it uses reading levels ("Beginning," "Intermediate," "Advanced Intermediate," and "Advanced"), rather than grade levels. By using reading levels, documents can be classified more broadly into reading groups, which makes this test ideal for SSL (Spanish as a second language) readers.

The FRASE graph is designed for most text and can be used for both primary and secondary-age reading materials, as well as SSL materials. It was trained on eight textbooks representing the four levels of difficulty:

- *Nueva Vista*
- *Español: A Descubrirlo*
- *Entre Nosotros*
- *Vista Hispánica*
- *Civilización y Cultura*
- *Multivista Cultura*
- *Español: Curso Primero*
- *Perspectivas*

These textbooks "covered a variety of reading topics which dealt with family life, entertainment, customs, the Hispanic people, history, and politics" (143). 300–350 word excerpts were randomly selected from each book, scored on a FRASE graph, and then objectively and subjectively validated.

These validations involved correlating the FRASE scores against "teacher judgments, Spaulding formula ratings, cloze test scores, and informal multiple choice tests" (144). (The "teacher judgments" validation involved 16 secondary-level Spanish teachers ranking the samples into the levels of beginner, intermediate, advanced intermediate, and advanced.) The comparisons were significantly correlated, with the following results:

- cloze tests: $r = -0.93$
- Spaulding scores: $r = 0.91$
- teacher judgments: $r = 0.95$
- multiple choice tests: $r = -0.97$

Below is an example of a FRASE graph:

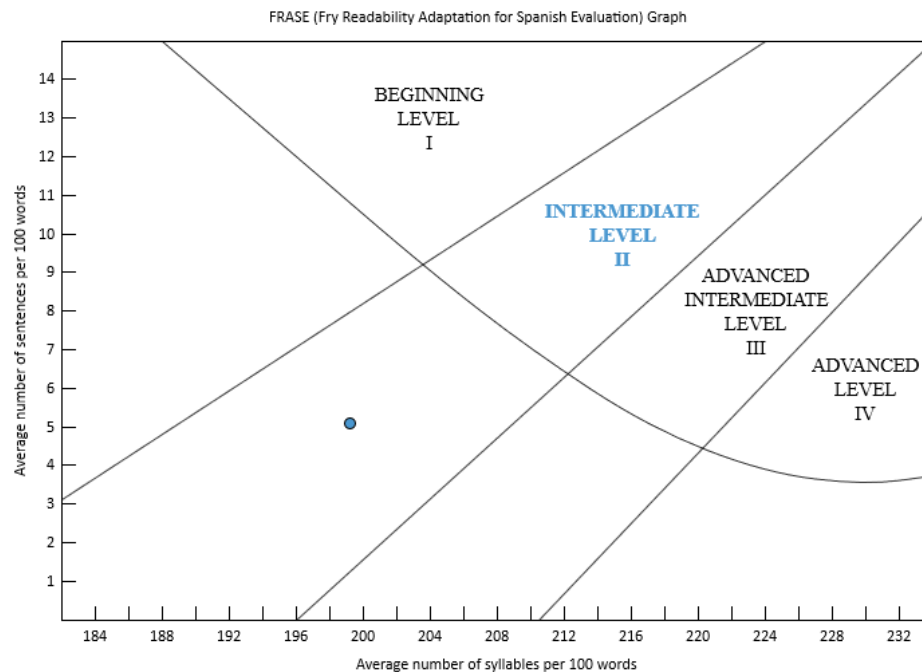


Figure 3.2: FRASE graph

The FRASE graph procedure is:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Count the number of syllables in each passage.
- Find the intersecting point on the plot. The quadrant where the point lies indicates the reading level of the document.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

3.3 Gilliam-Peña-Mountain Graph

Gilliam-Peña-Mountain (426–30) is a graphical test for Spanish text based on the Fry graph ("Clarifications, Validity, and Extension" 242–52). It is identical to the Fry graph, except that its X (syllable-count) axis begins at 175 (instead of 108). This is necessary to account for Spanish text having a much higher syllable count than its English counterpart. As the authors stated, "Spanish syllable counts, however, were so high that they threw many of the samples outside the bands or off the graph entirely" (427).

This test calculates the grade level from the average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

This variation of Fry was trained on samples from the following textbooks:

- *¡Vas bien!*
- *¿Ya te vas?*
- *¡Hasta luego!*
- *Nuestros amigos*
- *Senda 1*
- *Trabaja y aprende*
- *La ciudad*
- *Del campo al pueblo*
- *Aventuras maravillosas*
- *Otros amigos, otras culturas*
- *Senda 3*
- *Conozcamos a Puerto Rico*
- *Por tierras vecinas*

It was also trained on the following juvenile trade books:

- *Aquí viene el ponchado*
- *Osito*
- *El caso del forastero hambriento*
- *Leonardo el león y Ramón el ratón*
- *Jorge el curioso*
- *Teresita y las orugas*
- *Danielito y los dinosaurios*
- *Johnny Lost (Juanito Perdido)*
- *El niño que no creía en la primavera*

Syllable and sentence statistics were calculated from these samples. Initially, the researchers noted that most of these statistics could not be plotted onto a Fry graph, because of the high syllable count. They deduced that a constant would need to be subtracted from this syllable count to be able to plot it on the graphs. Next, they determined where on the graph these materials should lie based on either the grade level provided by the publisher or by plotting an English translation of the material onto a graph. By comparing the differences between the actual and expected syllable counts, it was determined that subtracting 67 from the Spanish syllable counts would adjust the score to appear on the graph where it was expected to be.

The Gilliam-Peña-Mountain graph is designed for both primary and secondary-age reading materials. Below is an example of a Gilliam-Peña-Mountain graph:

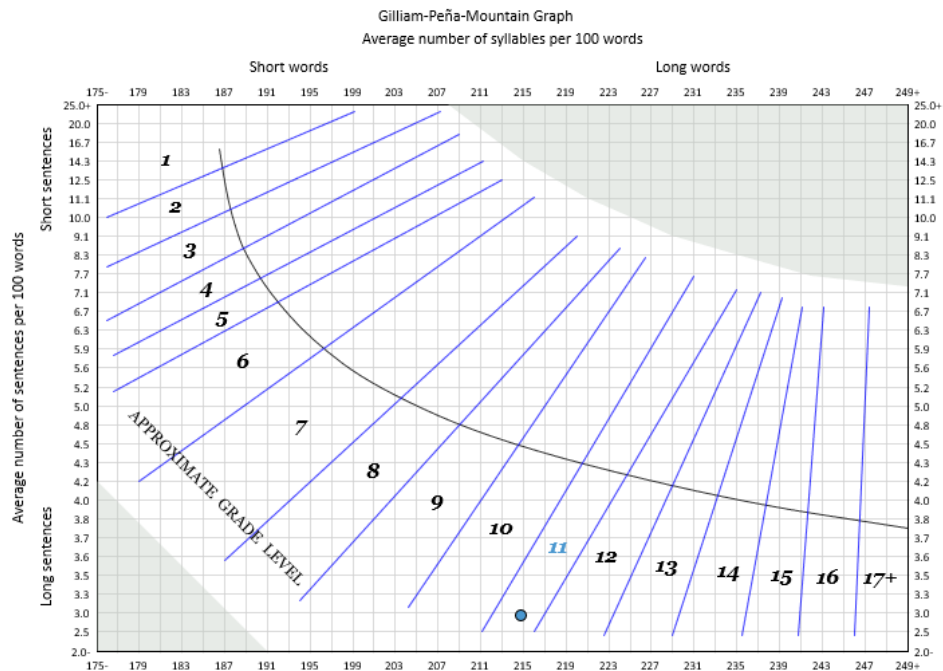


Figure 3.3: Gilliam-Peña-Mountain graph

The Gilliam-Peña-Mountain graph formula is:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Find the point on the chart.
- Count the number of syllables in each passage and subtract 67 to adjust for the higher syllabic nature of Spanish text. A preferred alternative method is to shift the syllable-count axis by 67, rather than adjusting the actual syllable count. This approach preserves the syllable count, while still being able to plot the score on the graph.
- If the sample's syllable or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 261 syllables per 100 words, then this will need to be adjusted to 249.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

Note that numerals are fully syllabized for this test. For example, "1945" will be counted as nine syllables ("u-no"- "nu-e-ve"- "cua-tro"- "cin-co").

3.4 INFLESZ Scale

The INFLESZ Scale is a Spanish test derived from earlier adaptations of Flesch Reading Ease (*Art of Readable Writing* 213; *Plain English* 23). It builds on Szigriszt's *Escala de Nivel de Perspicuidad*, which retained the original Flesch calculation, but adjusted its classifications for Spanish-language texts. While Szigriszt's Scale adjusted the difficulty levels for Spanish text, it relied on a relatively small and non-random corpus for validation. INFLESZ preserves Szigriszt's score, but realigned the scale boundaries and included fewer categories. This realignment was validated on a larger, more systematically sampled corpus. Hence, this revised scale is better suited for assessing the readability of a wide range of Spanish-language texts. (The authors' stated focus was on health-related materials.)

The INFLESZ Scale was validated using 210 Spanish-language publications drawn from three categories:

- *Publicaciones de quiosco* (newsstand publications), including national, regional, and sports newspapers, popular magazines, and comics
- *Textos escolares* (primary and secondary school textbooks) from major Spanish publishers
- *Revistas científicas* (scientific and medical journals aimed at health professionals)

From each publication, three random text fragments of at least 500 words were selected, yielding a total of 630 analyzed fragments.

Szigriszt index formula

$$I = 206.835 - (W/S) - (62.3 * (B/W))$$

Where:

I	Szigriszt index score
W	Number of words
B	Number of syllables
S	Number of sentences

Table 3.3: INFLESZ Scale Table

INFLESZ Score	Description
81–100	<i>Muy fácil</i> (Very easy)
66–80	<i>Bastante fácil</i> (Fairly easy)
56–65	<i>Normal</i> (Normal)
41–55	<i>Algo difícil</i> (Fairly difficult)
0–40	<i>Muy difícil</i> (Very difficult)

The INFLESZ category boundaries are based on the article's instructions that *Muy Fácil* applies to scores "puntuaciones superiores a 80" [scores greater than 80] (Barrio-Cantalejo et al. 140). This implies a strict greater-than cutoff for the upper boundary, rather than an inclusive threshold. As a

result, the INFLESZ cutoffs follow a different boundary logic than Flesch Reading Ease, which treats category limits as inclusive.

Below is an example of the INFLESZ Scale:

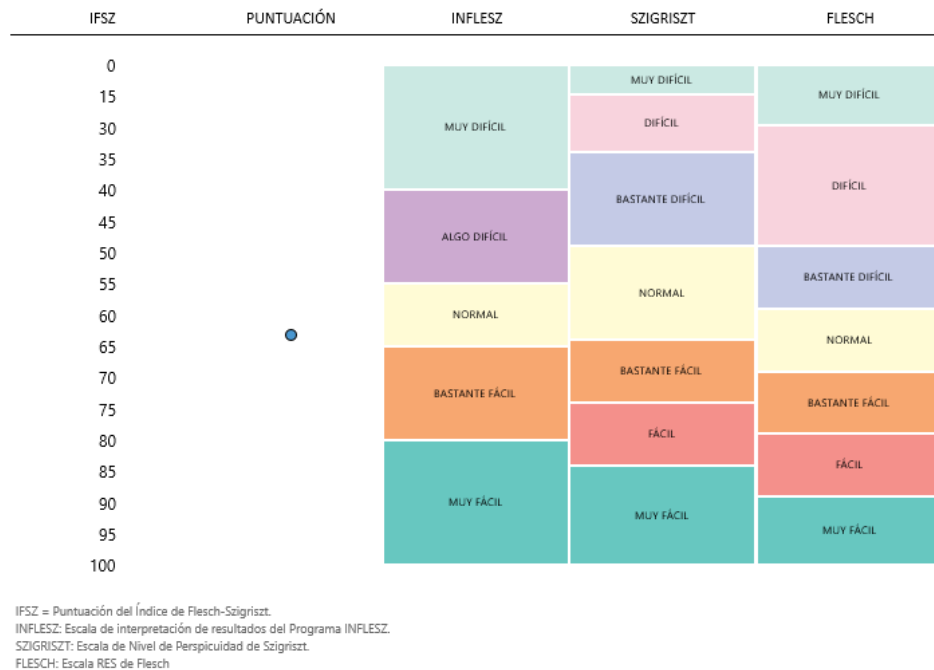


Figure 3.4: INFLESZ Scale

This chart consists of vertically aligned scales. The left scale, INFLESZ, presents which difficulty level the score falls into. (This is the primary focus of interpretation.) To the right are the original Szigriszt and Flesch Reading Ease scales, respectively. (These are only included for comparison.)

3.5 Lix

Refer to sec. 2.19.

3.6 Rix

Refer to sec. 2.31.

3.7 SOL

The SOL readability formula (Contreras et al. 21–29) calculates the grade level of a document based on complex-word density.

SOL is a modified version of SMOG (McLaughlin 639–46) that was recalibrated for Spanish text. This test calculates the SMOG score of a Spanish document, and then adjusts the score to take into account the comparably higher complex-word counts found in Spanish text. This adjustment is similar to what Gilliam-Peña-Mountain (Gilliam et al. 426–30) uses for plotting Spanish text onto the Fry graph.

This test was trained on multiple blocks of 30-sentence samples from the following English, Spanish, and French materials:

- *The Lancet* (Spanish and English)
- *El País* (Spanish)
- *Le Monde* (French)
- *International Herald Tribune* (English)
- *Don Quixote* (Spanish, English, and French)
- *The Bible* (Spanish, English, and French)
- *Snow White and the Seven Dwarfs* (Spanish, English, and French)
- *The Three Bears* (Spanish, English, and French)
- *The Wolf and the Seven Little Kids* (Spanish, English, and French)
- *The Three Little Pigs* (Spanish, English, and French)
- Story books (a set of story books) (Spanish, English, and French)

SMOG scores for these samples (including available translations) were collected. Next, the scores between each language for every set of material were correlated. The researchers found significantly higher SMOG scores for French compared to English, and higher SMOG scores for Spanish compared to French. To adjust for these differences, the following regression equation was built to convert a Spanish SMOG calculation to English ($r^2 = 0.60$):

SOL formula

$$G = (1.0430 * \sqrt{C} + 3.1291) * .74 - 2.51$$

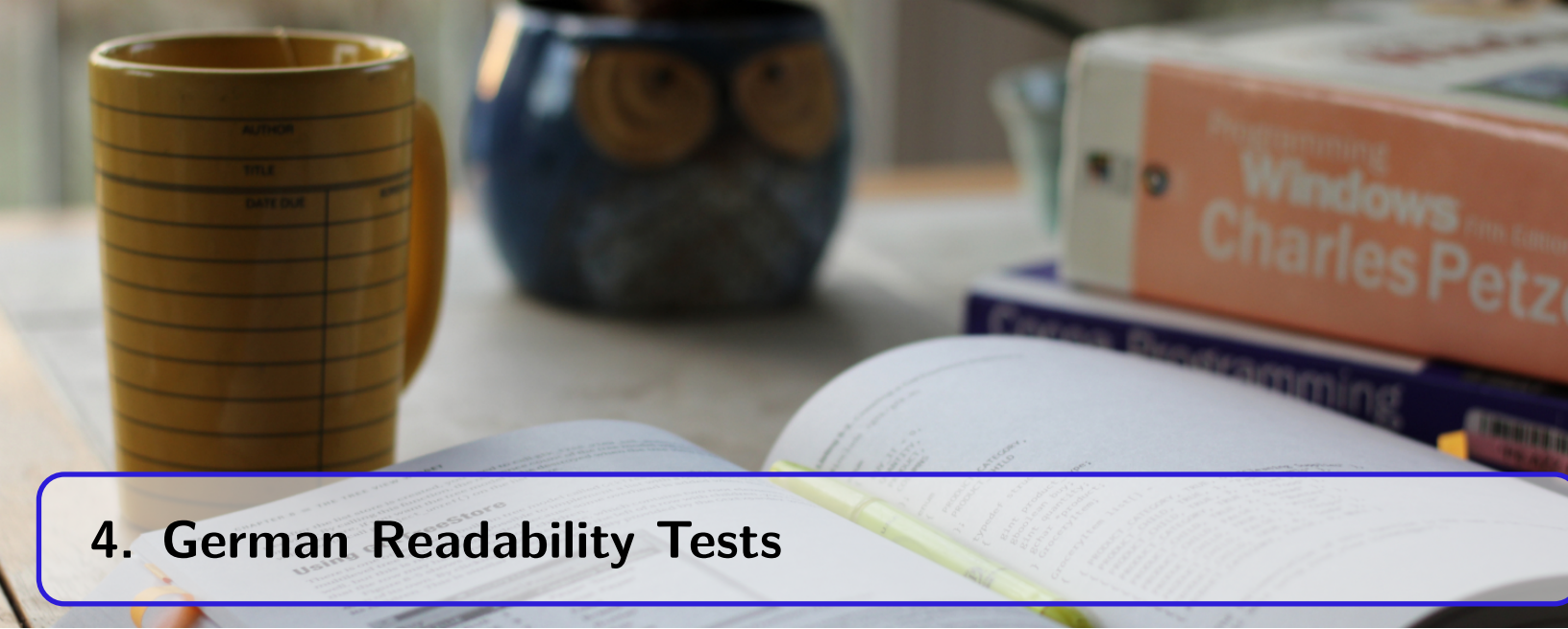
Where:

G	Grade level
C	Number of complex words (3+ syllables) from 30 sentences

Like SMOG, this test fully syllabizes numerals (i.e., sounds them out) . It also excludes lists and tables, but includes headers and footers (Contreras et al. 23). (SMOG does not explicitly follow this rule.)

Note that this test uses the high-precision version of the SMOG formula. This is recommended by the authors to avoid any rounding issues when converting scores between Spanish and English (24).

This test requires 3 ten-sentence samples, but normalization can be used to analyze larger samples or entire documents.



4. German Readability Tests

4.1 Amstad

The Amstad test is a recalculation of Flesch Reading Ease for German text (78–81, 50). Its formula is adjusted from the original Flesch to take into account German's comparatively longer words. It also uses a lower weighting for the sentence-length factor, placing most of the influence on word length.

The factors used for this test are syllables per word ("*Silben pro Wort*") and words per sentence ("*Wörter pro Satz*") (81). (These are the same factors used for Flesch Reading Ease).

Amstad is best suited for school textbooks and technical manuals. Scores range from 0–100 (the higher the score, the easier to read) and average documents should be within the range of 60–70.

Amstad formula

$$I = 180 - (W/S) - (58.5 * (B/W))$$

Where:

I	Amstad (i.e., Flesch) index score
W	Number of words
B	Number of syllables
S	Number of sentences

Amstad offered these interpretations for scoring:

Table 4.2: Amstad Conversion Table

Reading Ease Score	Description of Style	Typical Magazine
90–100	Very easy	comics
80–89	Easy	pulp-fiction
70–79	Fairly easy	slick-fiction
60–69	Standard	digest
50–59	Fairly difficult	quality
30–49	Difficult	academic
0–29	Very difficult	scientific

Note

Like Flesch Reading Ease, this test treats numerals as monosyllabic words by default. This behavior can be changed from the Options dialog.

4.2 Lix

Refer to sec. 2.19.

4.3 Lix (German children's literature)

This test is a German variation of Läsbarhetsindex (Lix) designed for children's literature. It uses the same index-score formula as Lix, but uses different scales for assigning the difficulty and grade-level scores. This adjustment was developed by Renström (152–153) and Bamberger & Vanecek (64, 187) for use with German literature ranging from 1st to 8th grade. (Renström developed the adjusted difficulty gauge; Bamberger & Vanecek developed the grade-level conversion.)

Note that this test differs from the original Lix approach of using the same scoring for any document, regardless of language. Instead, this test is designed specifically for German materials, adjusting the grade-level score and classification of the index value.

Regarding the index score, the following variation of the Lix gauge is used to classify the difficulty (Renström 153; Bamberger and Vanecek 65; Schulz 53):

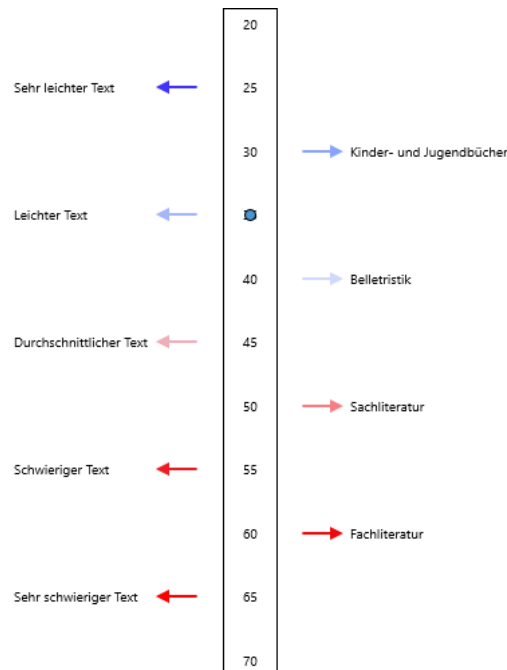


Figure 4.1: German Lix gauge (children's literature)

This version adjusts the scaling of the score, assuming that German text is inherently more difficult than other languages. It also includes additional levels of difficulty (e.g., "children and young adult").

To convert the Lix index score to a grade, this test uses the following table:

Table 4.3: Lix (German children's literature) Conversion Table

Index Score	Grade
0-23	1 st grade
24-26	2 nd grade
27-29	3 rd grade
30-31	4 th grade
32-33	5 th grade
34-35	6 th grade
36-37	7 th grade
38+	8 th grade

4.4 Lix (German technical literature)

This test is a German variation of Läsbarhetsindex (Lix) designed for technical literature. It uses the same index-score formula as Lix, but uses different scales for assigning the difficulty and grade-level scores. This adjustment was developed by Renström (152–153) and Bamberger & Vanecek (64, 187) for use with German literature ranging from 3rd to 15th grade. (Renström developed the adjusted difficulty gauge; Bamberger & Vanecek developed the grade-level conversion.)

Note that this test differs from the original Lix approach of using the same scoring for any document, regardless of language. Instead, this test is designed specifically for German materials, adjusting the grade-level score and classification of the index value.

Regarding the index score, the following variation of the Lix gauge is used to classify the difficulty (Renström 153; Bamberger and Vanecek 65; Schulz 53):

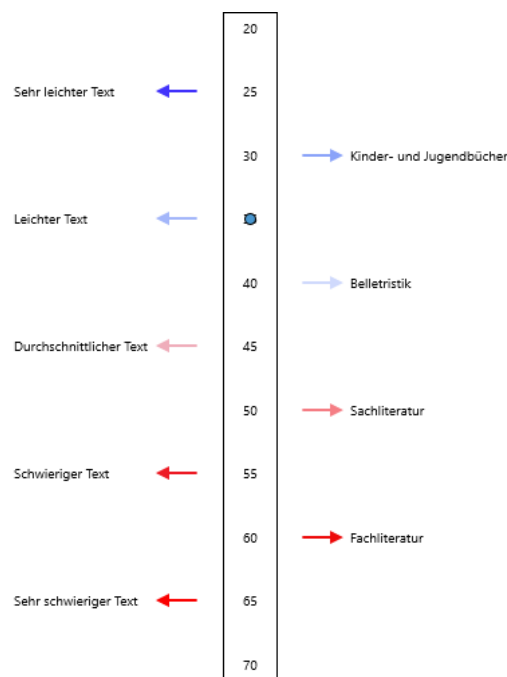


Figure 4.2: German Lix gauge (technical literature)

This version adjusts the scaling of the score, assuming that German text is inherently more difficult than other languages. It also includes additional levels of difficulty (e.g., “children and young adult”).

To convert the Lix index score to a grade, this test uses the following table:

Table 4.4: Lix (German technical literature) Conversion Table

Index Score	Grade
0-30	3 rd grade
31-33	4 th grade
34-37	5 th grade
38-40	6 th grade
41-43	7 th grade
44-47	8 th grade
48-50	9 th grade
51-53	10 th grade
54-56	11 th grade
57-59	12 th grade
60-63	13 th grade
64-69	14 th grade
70+	15 th grade

4.5 Neue Wiener Sachtextformel 1

Neue Wiener Sachtextformel 1 (1.nWS) (Bamberger and Vanecek 83, 187) is used for evaluating German nonfiction (specifically, 6th-10th grade materials). This test returns a grade-level score based on word length/complexity and sentence length.

Neue Wiener Sachtextformel 1 formula

$$G = 0.1935 * ((C/W) * 100) + 0.1672 * (W/S) + \\ 0.1297 * ((X/W) * 100) - 0.0327 * ((M/W) * 100) - 0.875$$

Where:

G	Grade level
C	Number of 3+ syllable ^a words
X	Number of 7+ character ^b words
M	Number of monosyllabic ^c words
S	Number of sentences
W	Number of words

^a"*drei- und mehrsilbigen*"

^b"*mehr als sechs buchstaben*"

^c"*einsilbigen*"

4.6 Neue Wiener Sachtextformel 2

Neue Wiener Sachtextformel 2 (2.nWS) was developed by Bamberger & Vanecek for evaluating German nonfiction (specifically, very light materials up to 5th grade) (84, 187). This test returns a grade-level score based on word length/complexity and sentence length.

Neue Wiener Sachtextformel 2 formula

$$G = 0.2007 * ((C/W) * 100) + 0.1682 * (W/S) + 0.1373 * ((X/W) * 100) - 2.779$$

Where:

G	Grade level
C	Number of 3+ syllable ^a words
X	Number of 7+ character ^b words
S	Number of sentences
W	Number of words

^a "drei- und mehrsilbigen"

^b "mehr als sechs buchstaben"

4.7 Neue Wiener Sachtextformel 3

Neue Wiener Sachtextformel 3 (3.nWS) was developed by Bamberger & Vanecek for evaluating German nonfiction (specifically, very light materials up to 5th grade) (84, 187). This test returns a grade-level score based on word complexity and sentence length.

Neue Wiener Sachtextformel 3 formula

$$G = 0.2963 * ((C/W) * 100) + 0.1905 * (W/S) - 1.1144$$

Where:

G	Grade level
C	Number of 3+ syllable ^a words
S	Number of sentences
W	Number of words

^a "drei- und mehrsilbigen"

4.8 Quadratwurzelverfahren

Quadratwurzelverfahren ("Square root method") is an alternative method for calculating SMOG (Bamberger-Vanecek) (78). Where SMOG (Bamberger-Vanecek) uses a 30-sentence sample, Quadratwurzelverfahren uses a 100-word sample; otherwise, they produce similar results.

Quadratwurzelverfahren formula

$$G = \sqrt{((C * (100/W)) / (S * (100/W)) * 30) - 2}$$

Where:

G	Grade level
C	Number of 3+ syllable ^a words
S	Number of sentences
W	Number of words

^a"*drei- und mehrsilbigen*"

Note

Based on the examples from Bamberger-Vanecek (78), the grade-level score should be truncated (or "floored") to one-point precision, not rounded.

4.9 Rix

Refer to sec. 2.31.

4.10 Rix (German fiction)

This test is a German variation of Rix designed for fiction books. It uses a modified index-score formula from Rix, as well as different logic for assigning a grade-level score. This test was developed by Bamberger & Vanecek (64, 187) for use with German fiction ranging from 1st to 11th grade.

Rix (German fiction) formula

$$I = ((100 * (X/W)) / (U * (100/W))) * 10$$

Where:

I	Rix index score
W	Number of words
X	Number of long words (7+ characters)
U	Number of sentence units

To convert the Rix index score to a grade, this test uses the following table:

Table 4.6: Rix Index Score (German Fiction) to Grade Level Conversion Table

Index Score	Grade
0-13.5	1 st grade
13.51-17	2 nd grade
17.1-20.5	3 rd grade
20.51-24	4 th grade
24.1-27.5	5 th grade
27.51-31	6 th grade
31.1-34.5	7 th grade
34.51-38	8 th grade
38.1-41.5	9 th grade
41.51-45	10 th grade
45.1+	11 th grade

4.11 Rix (German non-fiction)

This test is a German variation of Rix designed for non-fiction books. It uses a modified index-score formula from Rix, as well as different logic for assigning a grade-level score. This test was developed by Bamberger & Vanecek (64, 187) for use with German non-fiction ranging from 4th to 14th grade.

Rix (German non-fiction) formula

$$I = ((100 * (X/W)) / (U * (100/W))) * 10$$

Where:

I	Rix index score
W	Number of words
X	Number of long words (7+ characters)
U	Number of sentence units

To convert the Rix index score to a grade, this test uses the following table:

Table 4.8: Rix Index Score (German Non-fiction) to Grade Level Conversion Table

Index Score	Grade
0-26	4 th grade
26.1-32	5 th grade
32.1-38	6 th grade
38.1-45	7 th grade
45.1-52	8 th grade
52.1-57	9 th grade
57.1-66	10 th grade
66.1-75	11 th grade
75.1-84	12 th grade
84.1-100	13 th grade
100+	14 th grade

4.12 Schwartz German Readability Graph

The Schwartz German Readability Graph (1–21) is a derivative of Fry's readability graph and is designed for instructional textbooks for primary and secondary students. Like Fry, it relies on sentence length and word complexity (i.e., syllable counts) to determine a document's difficulty.

As an elementary teacher, Schwartz was familiar with English tests and had found them “extremely helpful in classifying texts according to levels and fitting a student with proper reading material” (3). Later, as a high school German teacher, she was often asked if there were any German readability tests.

After corresponding with Prof. George Klare and Dr. Richard Bamberger, she confirmed that only two German tests existed (as well as some unpublished cloze procedure experiments). She then interviewed West German publishers to see how they classified school textbooks. They informed her that their process was to compare student dictionaries against their materials to gauge difficulty, as well as use large print and illustrations.

Schwartz then set out to create a German readability test, given the lack of such a tool at the time. She initially experimented with word frequencies as a factor, but decided that sentence length and word complexity were easier to calculate when computer assistance wasn't available. She also determined that a graphical test—such as Fry—was easier to use by classroom teachers than regression formulas. Hence, Schwartz created a new graph similar to Fry, but adjusted it for the West German basal readers that she used in her study.

When designing the graph, Schwartz adjusted the syllable-count axis to take into account German words' tendency to have higher syllable counts. As she noted, “It is commonplace that the average German word contains more syllables than the average English one” (14). In her samples, she noted that English averaged 25 syllables per 100 words, while German averaged 37 (14).

Another difference from the Fry graph is that some grade bands cover two grades (whereas each band on a Fry graph represents a unique grade).

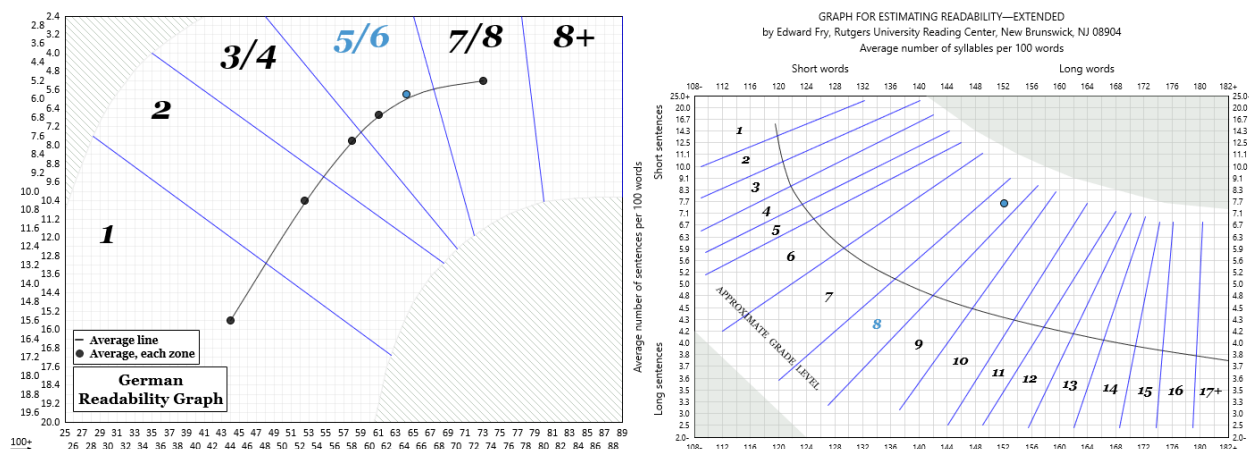


Figure 4.3: Comparison of bands between Schwartz and Fry graphs, respectively

As Schwartz noted:

Furthermore, it might be of interest to the American reader to note that it used to be quite common in Germany to have certain reader levels combined in one volume: usually third and fourth, fifth and sixth, seventh and eighth, sometimes even fifth through eighth, and that often there was no distinction made inside the book itself as to in which grade a particular text had to be read. (5–6)

In the study, between 15–21 random 100-word samples were taken from 2nd–8th grade textbooks and between 11–15 100-word samples from 1st grade textbooks. The following textbooks were used:

Grade 1:

- *Bunte Lesewelt*
- *Bei uns daheim*
- *Unsere neue Fibel*
- *Bunte Welt*
- *Mein erstes Buch zum Anschauen, Zeichnen, Lesen und Schreiben*

Grade 2:

- *Mein Lesebuch für das 2. Schuljahr*
- *Lesebuch für das 2. Schuljahr der Grundschule*
- *schwarz auf weiß, Primarstufe, Text 2*
- *Deutsches Lesebuch für die Grundschule für das zweite Schuljahr*
- *auswahl, Lesebuch für Schulen, Grundschule 2. Schuljahr*
- *Lesebuch für die Volksschulen, 2. Schuljahr*

Grade 3:

- *Mein Lesebuch für das 3. Schuljahr*
- *Lesebuch für das 3. Schuljahr der Grundschule*
- *Deutsches Lesebuch für das dritte und vierte Schuljahr*
- *schwarz auf weiß, ein neues Lesebuch- Drittes Schuljahr*

Grade 4:

- *Mein Lesebuch für das 4. Schuljahr*
- *Lesebuch für das 4. Schuljahr der Grundschule*
- *Geschichten Berichte Gedichte*
- *schwarz auf weiß, ein neues Lesebuch- Viertes Schuljahr*
- *Deutsches Lesebuch, Viertes Schuljahr*

Grades 5/6:

- *Kein schöner Land*
- *Bayerisches Lesebuch für das 5. und 6. Schuljahr*
- *Lesebuch für die Volksschulen*

Grades 7/8:

- *Bayerisches Lesebuch für das siebte und achte Schuljahr*
- *Lesebuch für die Volksschulen, Vierter Band. 7. und 8. Schuljahr*
- *Lesebuch. C 8 (8. Schuljahr)*

Schwartz noted the following validation results:

- Grade 1: 83% of the 70 samples fell in the grade 1 band on the graph, 91% falling in either the grade 1 or 2 bands.
- Grade 2: 30.5% of the 105 samples fell in grade 2, although 82% fell in the 1, 2, or 3/4 bands.
- Grades 3 and 4: 29% of the 164 samples fell in the 3/4 band, although 77% fell within the 2, 3/4, or 5/6 bands.
- Grades 5 and 6: 40% of the 72 samples fell in the 5/6 band, while 81% fell in the 3/4, 5/6, or 7/8 bands.

Altogether, of the 411 samples for grade levels 1–6, 75% of them either fell into the proper band on the graph, or within one grade level.

When counting the sentences and syllables in a document, Schwartz had the following recommendations:

- Exclude headers (“headlines”)
- Treat numerals as one syllable
- Speech set off by a colon mid-sentence should be counted as a separate sentence

Regarding the last point, it could be inferred that independent clauses set off by colons, semicolons, and em dashes should also be counted as separate sentences. Because of this, sentence units should be used when counting sentences. Numerals should also be counted as one syllable.

Plotting and interpreting the Schwartz graph is the same as other Fry-like graphs. Once you have found the average number of syllables and sentences per 100-words, plot the intersection of these two statistics on the graph. The band that the point falls in will represent the grade-level score of the sample.

 Note

An item of interest when interpreting this graph is the x-axis. Although the labels read 25, 26, etc., they actually represent 125, 126, etc. The 100+ with the arrow underneath it (to the left) indicates that the axis labels should have 100 added to them. This is done to make the labels more compact.

Although this is an unusual display, this is how the graph is drawn in the original article. Therefore, to preserve fidelity with the source material, it is recommended to use this same style.

4.13 SMOG (Bamberger-Vanecek)

SMOG (Bamberger-Vanecek) (78) is a German variation of SMOG designed for secondary-age readers. This test calculates the grade level of a document based on complex-word density.

Note that numerals are fully syllabized (i.e., sounded out) for this test.

SMOG (Bamberger-Vanecek) formula

$$G = \sqrt{C} - 2$$

Where:

G Grade level

C Number of 3+ syllable^a words from 30 sentences

^a"*drei- und mehrsilbigen*"

This test requires a 10-sentence sample, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

 Note

Based on the examples from Bamberger-Vanecek (78), the grade-level score should be truncated to one-point precision, not rounded.

4.14 Wheeler-Smith (Bamberger-Vanecek)

Wheeler-Smith (Bamberger-Vanecek) (77, 186) is a German variation of Wheeler-Smith designed for both primary and secondary-age reading material. It calculates the grade level of a document based on unit length and number of complex words.

Wheeler-Smith (Bamberger-Vanecek) formula

$$I = ((W/U) * ((C/W) * 100))/10$$

Where:

I	Index value
C	Number of 3+ syllable ^a words
W	Number of words
U	Number of units

^a"*drei- und mehrsilbigen*"

Then, find the index value in the following chart to acquire the final readability score:

Table 4.9: Wheeler-Smith (Bamberger-Vanecek) Conversion Table

Grade	Score
First grade	2.5–6
Second grade	6.1–9
Third grade	9.1–12
Fourth grade	12.1–16
Fifth grade	16.1–20
Sixth grade	20.1–24
Seventh grade	24.1–29
Eighth grade	29.1–34
Ninth grade	34.1–38
Tenth grade	38.1–42

Note

Although Bamberger and Vanecek (77, 186) do not specify using units instead of sentences, the use of units is assumed to match Wheeler-Smith (398).

Based on the examples from Bamberger and Vanecek (77, 186), index values should be truncated to one-point precision, not rounded.



5. Scoring Notes

5.1 Grade Level Results

Most readability tests that return a grade level include a mantissa (i.e., decimal value), which represents the month of the grade. For example, 1.6 would be the sixth month of first grade (Carver 306–07). However, this only applies to tests that are computed from a mathematical formula (generally, a regression equation). Tests that use different methods—such as lookup tables or graphs—usually only return a grade level as an integer.

For example, New Dale-Chall looks up your unfamiliar word and sentence counts from a table and returns a grade range. This will yield results such as 7–8, meaning that a document falls within the seventh to eighth grade reading level.

Lix and Rix use a similar method. These tests calculate an index score and then look up the grade level from a table using this value.

Finally, graphical tests (e.g., Fry) only return integral grade levels because the bands on these graphs only represent the grade, not the grade and month.

5.2 Cloze Scores and Readability Formulas

A cloze procedure is a test given to a student to measure their comprehension ability. This test consists of the following steps:

- Take a 250 word passage from a document.
- Delete every fifth word and replace it with a blank of equal length. There should be a total of 50 blanks when finished.
- Have the student fill in as many of the blanks as he/she can.
- Count the number of blanks that he/she filled in correctly. Traditionally, synonyms are not accepted, but this can be at the discretion of the administrator. Misspellings are acceptable.
- Multiply the number of correct answers by two. This is the percentage of correct answers and will range from 0 (zero comprehension) to 100 (total comprehension).

Cloze procedures can also be used to gauge how easily a passage of text can be comprehended. In this context, a cloze procedure is given to a group of students, and the mean (average) of their

scores is the level of comprehensibility for the passage. Sixty percent and higher are considered to be acceptable levels of comprehensibility. Scores lower than 60% indicate that the passage may be too difficult to be comprehensible and should be rewritten.

Cloze scores such as these are the criteria used to validate most readability formulas. Generally, researchers use the passages and results from published cloze procedures as their test data when creating a formula. This process usually involves multiple regression with the cloze scores of the passages as the dependent variable and factors such as sentence length and syllable count as the independent variables. This analysis will determine which factors most strongly affect the cloze scores. After finding the most effective factors, these factors are then incorporated into a formula. Through experimentation, the formula is adjusted to produce results that strongly correlate with the original cloze scores. The stronger this correlation is, the more accurate the formula will be in predicting the readability of new text samples.

Some readability formulas produce predicted cloze scores (rather than grade levels). A predicted cloze score represents an estimation of a passage's mean score if it were made into a cloze test and given to a group of students. This can be useful when selecting passages to use for cloze tests.

A photograph of a desk with a yellow mug, a blue owl mug, an open book, and a yellow highlighter. The yellow mug has a table with columns for 'AUTHOR', 'TITLE', 'DATE DUE', and 'PAGE'. The blue owl mug has large eyes. The open book has a yellow highlighter resting on it. The background is blurred.

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¹<https://www.edisonohio.edu/>

²<https://www.wright.edu/>



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